KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

INSTITUTE OF DISTANCE LEARNING

(BSC AGRICULTURAL ENGINEERING, THIRD YEAR)

AE 460: MAINTENANCE AND REPAIRS OF AGRICULTURAL MACHINERY

[Credit: 2]

MR. K. A. AIKINS

Publisher's Information

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COURSE OBJECTIVES

After going through this course, students should be able to:

1. Understand the basic concepts and types of maintenance practiced on agricultural machinery.
2. Understand the concept of planned maintenance and what it entails, and therefore be able to draw a planned maintenance scheme.
3. Appreciate the causes and effects of machine/ machine part failure.
4. Know what farm machinery repair entails and some basic repair operations carried out on machine parts.
5. Understand the spare parts supply chain and also be able to determine the economic order quantity of spare parts.
6. Appreciate the importance of workshop safety and means of creating a safe working environment.

COURSE OUTLINE

The course material has been structures into six (6) units.

- Unit 1: Basic Concepts of Maintenance Practices
- Unit 2: Planned Maintenance
- Unit 3: Troubles and Defects of Farm Machinery
- Unit 4: Repair Of Farm Machinery
- Unit 5: Basic Repair Operations
- Unit 6: Structural Connections
- Unit 7: Spare Parts
- Unit 8: Agricultural Workshop Safety

GRADING

Continuous assessment: 30%

End of semester examination: 70%

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Unit 1

BASIC CONCEPTS OF MAINTENANCE PRACTICES

Introduction

Unit 1 presents the basic concepts and definition of maintenance, types of maintenance and an introduction to maintenance probability and reliability.



Learning Objectives

After reading this unit you should be able to:

1. Understand what maintenance entails and why it is necessary.
2. Identify the various types of maintenance and their advantages and disadvantages.
3. Demonstrate a basic knowledge of maintenance probability and reliability.

Unit content

SESSION 1-1: MAINTENANCE

- 1-1.1: Reasons for machinery maintenance
- 1-1.2: Good maintenance practice
- 1-1.3: Maintenance Decision Making

SESSION 2-1: TYPES OF MAINTENANCE

- 2-1.1: Reactive maintenance
- 2-1.2: Conventional/ traditional maintenance
- 2-1.3: Proactive maintenance

SESSION 3-1: MAINTENANCE PROBABILITY AND RELIABILITY

- 3-1.1: Maintainability
- 3-1.2: Reliability

3-1.3: Availability

SESSION 1-1: MAINTENANCE

Maintenance can be defined as the practice of keeping of equipment, machine system or object in form or shape in its original state as much as possible. It is a combination of all technical, administrative and managerial actions during the life cycle of an item intended to retain or restore it to, a near-original state in which it can perform the required function (function or a combination of functions which are considered necessary to provide a given service). The old concept of maintenance was to attend to a machine after it has failed or stopped functioning. But now some more concepts have been identified and applied in the realm of ***maintenance.***

Regular maintenance is one of the prerequisites for a long lasting and reliable engine performance. The recommended engine servicing procedure for a particular system and engine is contained in the Owners Handbook for each engine model. ***Additional servicing*** is often necessary and depends on the type of operation the engine is subjected to.

1-1.1: Reasons for machinery maintenance

Thus the maintenance of these assets is necessary for the following reasons:

1. To preserve the service life of such machine.
2. To enhance machine performance at peak load i.e. to get the most of your machine.
3. To conserve cost in unnecessary procurement of new machine thereby increasing the overhead cost.
4. To restore machine to or near its original state at manufacture.
5. To increase output and profit
6. To provide the operator some comfort while in operation.

1-1.2: Good maintenance practice

Good maintenance practices are essential for efficient operation of all types of machinery. Efforts spent in this area of farm management is more than repaid by consistent and reliable operation of machinery, reduced fuel consumption and costs, extended machine life among others. Maintenance of farm machinery is complicated by the usage pattern of machine for short but intense period of activity, followed by long periods of non-use or storage.

To effectively safeguard the service life of your machine and guarantee its performance, you must protect it from the following enemies;

1. *Wear* (grease and oil are used to protect machines from wears),
2. *Dirt* (Filters are used to catch and hold dirt before it gets inside and damages parts),
3. *Heat* (the cooling and lubrication systems protects the machine from heat).

1-1.2.1: Objectives of good maintenance practices

Objectives of a good maintenance practice include the following:

1. To intervene before failure occurs
2. To carry out maintenance only when required and not when necessary.
3. To reduce number of failure and shutdown times.
4. To reduce maintenance cost and breakdown cost due to production lost.
5. Increase service life of equipment.
6. Reduction in inventory cost / effective inventory control.

During the "standing" or non-use periods, chemical interactions between metals and fluids can cause more damage than normal wear and tear from active usage. This must be considered in planning machinery maintenance and the following suggestions are worthy of consideration.

i. Follow manufacturers' instructions for all settings, adjustments, maintenance instructions, operating requirements and long term storage.
ii. Follow manufacturers' recommendations on safety aspects of operation and repair.
iii. Maintain all safety equipment as installed or recommended by manufacturers.
iv. Do not overload equipment, or operate at higher speeds than manufacturer recommends.
v. Do not add counterweights to equipment to increase load capacity unless authorized by manufacturer. Store equipment in clean and dry conditions.
vi. Remove all vegetation such as grass, hay, crops and crop residue from equipment before storage periods. Decomposing vegetable matter causes corrosion to metal surfaces. This is particularly important where surfaces are polished from usage.
vii. Keep all cutting edges sharp and clean. Sharp cutters require less power and reduce overall load on equipment. Cracked or damaged cutting edges are also easier to detect on clean equipment.
viii. Replace these items at end of the season rather than at season commencement.
ix. Inspect machinery at the end of season or harvest.

x. Repair and adjust as required. Carry out maintenance work without pressure between seasons.

1-1.3: Maintenance Decision Making

Decision makers need to take into account the following maintenance priorities:

1. The ***needs*** (interests) of their business,
2. ***Recommendations*** from the original equipment manufacturer,
3. Their own ***experience*** and that of other users of similar equipment, and
4. Online or offline ***information*** on available machine conditions.

To establish and implement any maintenance action there is the need to take decisions on:

1. Whether maintenance personnel are provided by the organization or whether they are obtained from external sources;
2. Who provides spare parts, materials and consumables, e.g. inventory, local sourcing or external supply;
3. Where special tools, transportation, lifting, testing and support equipment is sourced;
4. Condition monitoring equipment and software to be used;
5. Infrastructure that needs to be provided to implement maintenance policies.



Self-Assessment 1-1

1. Briefly explain the term maintenance.
2. Why is machinery maintenance necessary?
3. To effectively safeguard the service life of an engine and guarantee performance, one must protect it from three enemies?
    a. What are these enemies?
    b. How is the engine protected from these enemies?
4. What are the objectives of good maintenance practices?
5. Which suggestions are worth considering during the non-use periods of farm machinery to keep them safe from chemical interactions between metals and fluids which can cause more damage than normal wear and tear from active usage?

SESSION 2-1: TYPES OF MAINTENANCE

Many types of maintenance techniques have been used in machinery and equipment maintenance and reconditioning and are classified as follows: ***reactive maintenance*** (corrective or break down maintenance), ***conventional maintenance*** (which include routine maintenance and preventive maintenance) and ***proactive maintenance,*** etc.

Traditionally, maintenance is performed in either time-based-approach on fixed intervals, called preventive maintenance, or by failure-based-approach termed corrective maintenance. For some equipment and faults, corrective maintenance action must be performed immediately, while for others the maintenance action can be deferred in time, all depending on the equipment function.

The different types of maintenance are discussed as follows;

2-1.1: Reactive maintenance

Reactive maintenance is basically the "run it till it breaks" maintenance mode. No actions or efforts are taken to maintain the equipment as the designer originally intended to ensure design life is reached.

Advantages to reactive maintenance can be viewed as a double-edged sword. If we are dealing with new equipment, we can expect minimal incidents of failure. If our maintenance programme is purely reactive, we will not expend manpower or incur capital cost until something breaks.

Advantages

- Low cost involvement
- Less staff required.

Disadvantages

- Increased cost due to unplanned downtime of equipment
- Increased labour cost, especially if overtime is needed.
- Cost involved with repair or replacement of equipment.
- Possible secondary equipment or process damage from equipment failure.
- Inefficient use of staff resources

2-1.2: Conventional/ traditional maintenance

2-1.2.1: Routine maintenance

Routine maintenance can be defined as maintenance actions carried out at regular intervals, i.e. on a time- or machine-run-based schedule. It involves periodic checks of relevant areas. The frequency of such checks ranges between hourly, daily, weekly and monthly or as recommended by manufacturers. Readings obtained from such checks could be collated in a maintenance record over a long period to give a behaviour history of the equipment. Routine maintenance reduces fuel bills and extends equipment life.

Activities such as changing lubricant are based on time, like calendar time or equipment run time. Tractor usage is measured in hours, not miles. For example, some tractor owners/ operators change their engine oil after every 100 hours of run time. No concern is given to the actual condition and performance capability of the oil. It is changed because it is time. Routine maintenance schedule are normally given in a machine's operators'/ owner's manual. For a tractor, for instance, it would include chassis lubrication, engine, transmission, and hydraulic oil change, filter changes, etc.

2-1.2.2: Preventive maintenance

This is one of the oldest and traditional methods of maintenance. Preventive maintenance is predetermined work performed on a machine with the aim of preventing the wear and tear or sudden failure of equipment components and to reduce rate of depreciation. It also keeps time loss due to breakdown to a minimum. Examples are washing and cleaning, change of oil, topping of battery electrolyte, lubrication, inspection and minor adjustments of pressure, flow, tightness of belts and chains, etc. All these are carried out with the aim of preventing machine conditions from going beyond safe limits.

Mechanical, process or control equipment failure can have adverse results in both human and economic terms. In addition to down time and the costs involved to repair and/or replace equipment parts or components, there is the risk of injury to operators, and of acute exposures to chemical and/ or physical agents. Preventive maintenance, therefore, is a very important ongoing accident prevention activity, which is integrated into operations/ product manufacturing process.

Advantages

1. Cost effective in many capital intensive processes.

2. Flexibility allows for the adjustment of maintenance periodicity.
3. Increased component life cycle.
4. Energy savings.
5. Reduced equipment or process failure.
6. Estimated 12% to 18% cost savings over reactive maintenance programme.

Disadvantages

1. Catastrophic failures still likely to occur.
2. Labour intensive.
3. Includes performance of unneeded maintenance.
4. Potential for incidental damage to components in conducting unneeded maintenance.

2-1.2.3: Opportunistic maintenance

This maintenance arises when an equipment or system is taken down for maintenance of one or two components, the opportunity can be utilized for maintaining other wearing out components. In multi component systems with several failing components, it is often required to use opportunistic maintenance. This would be probably economically. It is also useful in non-monitored components.

2-1.3: Proactive maintenance

Proactive maintenance is a type of maintenance that is employed as a response primarily to results from equipment assessment and predictive actions aimed at solving machine problems from the source before they become visible. With proactive maintenance, a closer look on machine damage by concentrating on the causes instead of the symptoms is made. It involves in-situ measurements that reveal the actual conditions of a machine/ equipment in operation. This is done to detect the onset of system degradation (lower functional state), thereby allowing causal stressors to be eliminated or controlled prior to any significant deterioration in the physical state of the component. Thus, proactive maintenance is both predictive and preventive. It manages trend values, by measuring and analysing data about deterioration and could employ a surveillance system, designed to monitor conditions through an on-line system. Need for maintenance is based on the actual condition of the machine rather than on some pre-set schedule.

Proactive maintenance is the single most important means of achieving savings unsurpassed by conventional maintenance. Imagine being able to pinpoint and eliminate a disease long

before any symptoms occur in your body. It would save you money spent on doctor's bills and keep you out of the hospital in the long run. Therefore, proactive maintenance may include strict contamination control programme for lubrication fluids, hydraulic fluids, gear oils, and transmission fluids. Examples are the use of air cleaners, fuel filters, oil filters, etc



Self-Assessment 1-2

1. Differentiate between reactive, conventional and proactive maintenance.
2. Briefly explain the following types of maintenance.
    a. Reactive maintenance.
    b. Routine maintenance.
    c. Preventive maintenance.
    d. Opportunistic maintenance.
    e. Proactive maintenance.
3. State the advantages and disadvantages of reactive maintenance.
4. State the advantages and disadvantages of preventive maintenance.

SESSION 3-1: MAINTENANCE PROBABILITY AND RELIABILITY

3-1.1: Maintainability

Maintainability is the probability that a failed component or system will be restored or repaired to a specified condition within a specified period of time when maintenance is performed in accordance with prescribed procedures. It gives a measure of the ease and rapidity with which a system or equipment can be restored to operational status following a failure. The following time related functions/ quantifiable measures are defined with respect to maintenance and repairs.

Mean time to repair (MTTR) an assembly: This includes both the time elapsing before it is realized that there has been a failure, which will be very short for self-revealing faults and the

time for maintenance personnel to be deployed and achieve their repair mission. Good design can reduce MTTR and it is also influenced by maintenance procedure.

Mean time to failure (MTTF): This is the time taken for failure to occur after repairs have been carried out. Good design can increase MTTF.

Mean time between failures (MTBF): This is the measure of the recurrence of failure in a system. It is the mean time between two successive failures and is the sum of MTTF and MTTR i.e.

$$MTBF = MTTF + MTTR$$

3-1.2: Reliability

Reliability (r) is a machine's ability to perform its functions under specified conditions without departing from the preset operating parameters. It shows the probability of survival of any system or machine component(s) throughout the time the machine is in operation. MTTF is one of the characteristics of reliability.

$$R(t) = e^{-\lambda t}$$

Where λ = Constant failure rate per second, i.e. Failure/ time.

With an unrealistic but ideal assumption of a constant repair rate, the situation can be analysed with reference to reliability and failure. Maintainability is mathematically expressed as:

$$\text{Maintainability (in time } t) = 1 - e^{-t/MTTR}$$

3-1.3: Availability

The concept of availability is the fraction of time for which a machine or component(s) can perform its desired/designed functions.

$$\text{Availability} = \frac{MTTF}{MTTF + MTTR} = \frac{MTTF}{MTBF}$$



Self-Assessment 1-3

1. Define the following terms:
    a. MTTR
    b. MTTF

c. MTBF
d. Reliability
e. Availability

Unit 2

PLANNED MAINTENANCE

Introduction

Unit 1 introduces students to the concept of planned maintenance and the benefits associated to its implementation in any workshop or organisation. Various aspects of planned maintenance such as the four basic questions of maintenance and how they are systematically answered to produce a planned maintenance scheme is discussed.



Learning Objectives

After reading this unit you should be able to:

1. Appreciate the concept of planned maintenance and its benefits.
2. Follow a systematic approach to draw a planned maintenance scheme.

Unit content

SESSION 1-2: DEFINITION OF PLANNED MAINTENANCE

1-2.1: Benefits of Planned Maintenance

SESSION 2-2: BASIC MECHANICS OF PLANNED MAINTENANCE

2-2.1: Initiating the scheme

2-2.2: The Inventory

2-2.3: Maintenance schedule

2-2.4: Job specification

2-2.5: The maintenance programme

2-2.6: Job Report

2-2.7: The history record

SESSION 1-2: DEFINITION OF PLANNED MAINTENANCE

Planned maintenance is the application of maintenance tackled in a scientific manner. It is not itself a specific type of maintenance. It is characterised by the comprehensive planning of maintenance operation. It gives the owners and the property managers more time to prepare for the works and, more importantly, to secure the necessary funding. Thus, one can say that planned maintenance is work organised and carried out with forethought, control and records and it applies equally to any forms of maintenance (preventive, corrective and proactive). For a successful implementation of planned maintenance:

i. The maintenance policy must be carefully considered
ii. The application of the policy must be planned in advance
iii. The work must be controlled and directed to conform to the pre-arranged plan
iv. Historical and statistical records must be compiled and maintained to help assess the results and to provide a guide for future policy

As stated previously, the ultimate purpose of industry is to produce goods at the most economical cost. Therefore, the maintenance plan must be organized to achieve this objective. It must consider the overall situation and embrace not only the maintenance and production functions but also the economics. Initially, each department-maintenance, production and accounts- will have its own pre-conceived ideas on what constitutes the best maintenance plan; but when the objective is defined and the problem studied on a combined basis, it is possible to develop a unified maintenance plan that will give the best overall results.

Where a policy of planned replacement is practised, the planning is mainly of a financial nature - ensuring money is available for the purchase of new equipment, balancing depreciation costs against tax allowances, negotiating the sale, or trade in, of old equipment at the most favourable price - but by far the greater part of planned maintenance is concerned with the organization of the engineering aspect of maintenance and us relationship to production.

This maintenance may be effected in different ways; when the facility is running, when it has stopped, before it breaks down, after it breaks down, it can be planned or it can be done as and when the need arises without any prior planning.

The fundamental basis of any planned maintenance system is deciding in advance:

- The individual items of plant and equipment to be maintained.

- The form, method and details of how each item is to be maintained
- The tools, spare parts and personnel and time that will be required to carry out this maintenance.
- The frequency at which these maintenance operations must be carried out.
- The method of administering the system.
- The method of analysing the results.

Expressing these basic essentials in a manner that will form the structure of a practical system; there must be:

1. A schedule of all the plant and equipment to be maintained
2. A complete schedule of all the individual tasks that must be carried out on each item of plant
3. A programme of events indicating when each task must be carried out
4. A method of ensuring that the work listed in the programme is carried out
5. A method of recording the results and assessing the effectiveness of the programme.

Even the simplest of planned maintenance schemes must be controlled, and certain facts need to be readily available to the person in control if this is to be effective. As the human memory cannot retain more than a very limited amount of information with any guarantee of accuracy, and as the scribbling of notes on odd scraps of paper - which are often lost-is equally useless and unreliable, a well-organized, more permanent system of documentation is essential.

There are many basic documentation systems available that have been designed specifically for the control and support of planned malignance schemes, and they are all capable of being tailored to suit the individual requirements of any company or organization. Frequently, these basic systems are elaborated to provide information on costs, man hours, repair times, materials and replacements used or any other relevant data that are considered helpful. But any system that is installed should be easy to operate, involve the maintenance staff in the minimum of paper handling and recording but must be capable of indicating clearly:

- what is to be maintained,
- how it is to be maintained,
- when *it is to be* maintained and
- is *the maintenance* effective.

1-2.1: Benefits of Planned Maintenance

Any company planning to adopt planned maintenance will of necessity want to know:

- Whether it can be successfully applied in their particular case and also
- What benefits will result?

Even though planned maintenance can be applied successfully to all types of industry, it ultimate effects and benefits will vary with respect to the industry, the local conditions, the depth and scope, and the form of application.

It should be noted that planned maintenance is not a panacea for every maintenance problem. It will not compensate for poor workmanship, lack of tools, bad design or maloperation of machinery by personnel. It will not convert worn-out obsolete equipment into modern, highly efficient units. However, since maintenance is carried in a more controlled, systematic and constructive manner, it leads to benefits as follows.

1. Greater availability of equipment or plant due to:
    a. Fewer breakdowns
    b. Maintenance carried out at most convenient time
    c. Less downtime than infrequent ad hoc maintenance
    d. Reduced length of downtime, as spare parts and equipment are known in advance and are available when needed
2. Cheaper servicing (as those servicing are regular & simpler) than sudden expensive stop gap repairs.
3. Continuously high level of operational output, quality, performance and efficiency.
4. Greater and more effective labour utilization in the sense that:
    a. The volume of maintenance work is planned so that it is spread evenly throughout the year, and widely fluctuating demands on labour force is reduced.
    b. The weekly workload is known in advance so that its allocation can be controlled effectively.
    c. The effect of a positive maintenance policy is frequently reflected in an improved personal attitude of the staff. A purposeful approach – a high morale.
5. Servicing and adjustments times as contained in the programme are not overlooked or omitted.

A very important aspect of any planned maintenance system is the collection, recording and subsequent interpretation of maintenance data. This does not only help assess the result to assist forward planning but also offers management other benefits such as:

- Improved budgetary control – realistic budgets can be formulated and subsequently controlled.
- Improved stock control of spare parts – realistic quantities of spare parts are stocked and re-ordered.
- Provision of information upon which management can make realistic forecasts and decisions
- Focusing attention on frequently recurring jobs and types of defect, and the type, frequency and cost of individual repairs



Self-Assessment 2-1

1. Briefly explain the concept of planned maintenance.
2. Which decisions are to be made in advance as fundamental basis of any planned maintenance system?
3. What are the four broad questions one would have to answer first in drawing a planned maintenance scheme?
4. What are the broad benefits of planned maintenance?

SESSION 2-2: BASIC MECHANICS OF PLANNED MAINTENANCE

2-2.1: Initiating the scheme

Usually, planned maintenance schemes are initiated from a higher management level. However, this level does not formulate the exact details or operation procedures but must define clearly the maintenance objectives and the framework of operation. Management appoints a person to whom the task is delegated. This person could be the works engineer, chairman of maintenance committee. This person:

1. Interprets the Management policy in a manner intended
2. Must be given:
    - A comprehensive brief about the policy
    - The boundaries within which he can operate
    - The results he is expected to achieve
    - A programme of target dates
    - The resources at his disposal
3. Must be given, within the limits of his brief:
    - Authority to make decisions
    - Power to carry out the decision
    - Responsibility for the actions taken

Such an appointment as well as the proposed maintenance procedures may initially create speculations, resentment and resistance from staff. Therefore, necessary steps must be taken initially to avoid these problems. The person appointed must:

1. Consult and inform staff at the very beginning
2. Keep staff in the picture as the scheme progresses
3. Note the timing and method of his introduction
4. Watch his own personal approach.

This person on appointment must:

1. Make general appraisal of the situation
2. Survey the task in hand and available resources
3. Work out a tentative approach to enable him apply the planned maintenance procedures in a manner that suits the system

Even if he is already a staff, the above tasks could enable him to have a fresh look, conceive new ideas and solutions for old familiar problems.

2-2.2: The Inventory

Inventory is a list of all facilities made for the purpose of identification. This clarifies and answers the question "*what is to be maintained?*" This can sometimes be obtained from readily available source, if records have already been kept. However, the records may require some corrections, as some additions or deletions may be needed because some replacements and movements of equipment/ items may not have been previously recorded.

An inventory sheet must be prepared with additional columns to allow for additions. The sheet normally has:

1. *Identification symbol* – Each item must have a positive means of identification.
2. *Description of Facility* – A brief description of the items concerned (sometimes with name plates).
3. *Location* – The department, section or area, in which the item is located, e.g. machine shop, boiler house, compressor room.
4. *Type* – e.g. Mechanical, electrical or sometimes with more detailed classification
5. *Priority rating* – Indicating the relative importance of each item.

2-2.2.1: Identification of Equipment

After inventory, it becomes essential to identify each item positively. The main reason for this identification in summary are:

1. So that there can be no mistake about which particular item is being referred to
2. To identify and relate the respective items with their relevant documents, and vice-versa
3. To locate the whereabouts of the respective items within the site
4. To indicate the department, section, group or type of item for cost allocation purposes

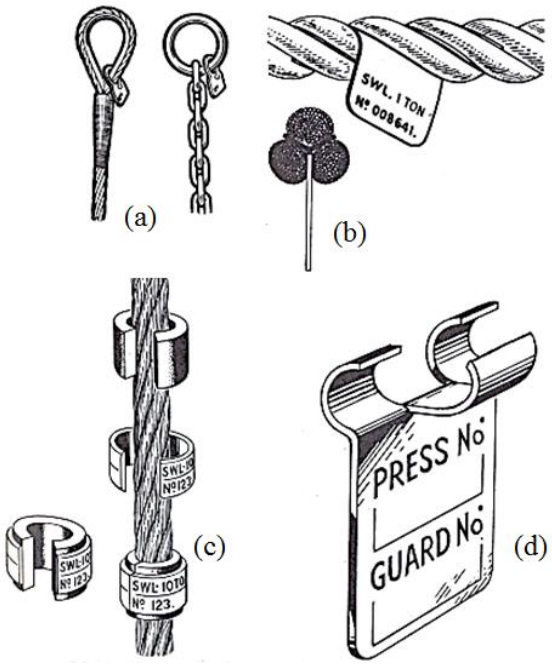


Figure 2-1: Identification labels

(a) Chain tabs, (b) Ropetap, (c) Cabletabs and (d) Guardtab

Identification is done using codes made up of colours, shapes, patterns, names, letters, numbers, or a combination of any of the above. Although marking of equipment can be effected in several ways as shown in Figure 2-1, the following features must not be compromised. The marks must:

1. Be easily visible (distinctive) and standardised where possible.
2. Be permanent (indelible).
3. Not impair or stress the item or the part to which it is attached.
4. Not cause obstruction, inconvenience or any hazard
5. Be cheap and easily obtainable.
6. Easily and quickly fixed.
7. Be neat in appearance.

2-2.2.2: The Facility Register

The facility register is a record of facilities (machinery and equipment), including their technical details. Items to be included in the maintenance plan should be selected from the inventory and the vital statistics about each entered in the facility register. The register records in a convenient and concise form all the essential details about each item so that the information is readily available as standard reference.

The information from the facility register on each item:

- Normally confirms the original specification, performance and manufacturers recommended limits, fits, tolerance.
- Helps in the ordering of correct spare and replacement parts.
- Provides the necessary information for planning the movement, relocation, access, safe floor loading, and layout of the workshop/unit

Before compiling the register, it is necessary for one to decide on:

- What information is to be recorded,
- How the information will be recorded,
- How the recorded information will be stored, and
- How the register will be formed.

Registers have for some time been made of sheets or cards, normally one card or sheet for each item. Each sheet or card must bear the identification symbol of the item or equipment

being referred to and they are filed together into books and the cards are kept together as flat trays.

Limitations to the use of sheets or cards are as follows:

1. Large spaces are required, especially in large firms
2. Retrieval of information has to be done manually
3. This at times, can be very time-consuming, daunting, tedious
4. It is also subject to mistakes

The activities with the facility register could be better enhanced by a fully computerised data storage and retrieval system. This occupies less space but is somehow costly, and one has to be trained on the Software of the system, with additional cost implications.

2-2.3: Maintenance schedule

Maintenance schedule is a comprehensive list of maintenance and its incidence. This answers the question "*how the equipment is to be maintained?*" For each facility included in the plan, a separate maintenance schedule must be compiled which set out all the tasks, which may include inspection, lubrication, adjustments, component replacement and overhaul together with the frequencies that are considered necessary to maintain the facility efficiently. The maintenance policy for each facility/ equipment should take into consideration its function and influence in the production process.

Maintenance schedule for each equipment consist of an individual sheet, card or a set of sheets/ cards that indicate:

1. Name and identification number of equipment
2. Location of the item
3. Reference no. of the schedule
4. Safety procedures to be followed
5. Detailed list of tasks to be carried out
6. Frequency at which each task must be carried out
7. Tradesmen/personnel required to undertake the tasks
8. Time required to carry out the task
9. Special tools or equipment required
10. Materials, major components or replacements required

11. When the facility is available for maintenance (running maintenance, shut down maintenance).
12. Associated equipment or facilities that should be maintained simultaneously
13. Details of any contract maintenance

From the completed schedules, information should be extracted and acted upon. This ensures that spare parts, replacements, tools, special equipment and other necessary items could be ordered on time. Also, maintenance contractors could be contacted and all the necessary pre-arrangements made and confirmed so that all parties know what is expected of them and are in agreement

2-2.4: Job specification

This is a document that describes the work to be done. It is a means of communicating the work details to the persons who will effect the work (i.e. a link between the planner and the doer). Each task listed in the maintenance schedule should have its own separate document(s) specifying the work to be done. The most elementary form of job specification is a direct extract from the maintenance schedule. A simple example of this is "withdraw and clean pistons, check that rings are free and the wear is within the recommended tolerance". Information could also be obtained from the manufacturer's manual, otherwise, one has to rely on previous experience, personal judgement, etc. Job specification is done basically to ensure that:

1. The task or job is carried out in the manner intended
2. The possibility of omitting an operation is minimized
3. Acceptable limits of wear, etc. and tolerances are clearly defined
4. The tradesman or operator knows the work and how it is to be done
5. The operation is always carried out in the same manner and the same allowances are always applied. Checking for wear, etc., is carried out in the same position and in the same way. Results over the course of time are all comparable (standardisation).
6. All persons doing the work, even for the first time, follow the same procedure (continuity).
7. A reference standard is available with which the job and work content can be compared to detect any variation

What to include in a job specification include:

1. The identification number and name of the item of equipment.
2. The location of the item.
3. The maintenance schedule reference number (and task reference) from which the task is extracted.
4. The job specification reference no. Each job specification must have its own reference number.
5. The frequency at which that particular task must be carried out
6. The trade or trades required to carry out each operation
7. The specific details of the work to be done - The procedure and method, details of inspections, tests, servicing, etc., allowable limits and tolerances.
8. Components to be replaced.
9. Special tools and equipment to be used.
10. Reference drawings, manuals, etc., applicable.
11. Safety procedures to be followed.

There should also be a footnote to the effect that:

Any minor defect observed in the course of the specified work but not listed for attention should, if possible, be corrected and reported. If it is not possible or convenient to correct the defect in the course of the specified work it should still be reported. If the defect is considered sufficiently serious it should be brought immediately to the attention of the authorities.

2-2.5: The maintenance programme

This is a list allocating specific maintenance to a specific period. Having established what is to be maintain, how it is to be maintained, it must now be determined when (or within what period) it is to be maintained. The main purposes of the maintenance programme are:

1. To set out a plan of work
    a. To spread the maintenance work load evenly over the year
    b. To ensure that no facility or maintenance task is omitted
    c. To ensure that the required maintenance is carried out at the specified frequency
    d. To co-ordinate the maintenance of associated facilities
    e. To co-ordinate the maintenance with operational requirements

2. To present an overall picture of maintenance work, present and future commitments (on short and long term basis). To assist forward planning, ordering of spare parts, future labour requirements, basis for budgetary control.
3. To act as a reminder of future maintenance events (on a short term basis)
    a. To formulate weekly work plans (for immediate future)
    b. To arrange for availability of equipment for operation
    c. To arrange or check available of labour, spare parts, sub-contractors, etc.

The maintenance programme has flexible time span, usually on an annual basis. The programme is normally prepared in consultation with the operators who are able to advice on schedules of operations. The programme must ensure that the planned downtime for the shutdown maintenance is considerably less than the downtime previously experienced. A time allowance must be included for unexpected contingencies. The plan must also allow for quick modification

2-2.5.1: Planning Charts

These are chats with time scale, usually in days or weeks, marked along the top edge. It depicts the work in progress and future commitments. It also presents various maintenance programmes with colour coded pins, pegs or crayons. The required maintenance and maintenance intervals are entered in appropriate columns (with allowance for holiday periods). It is to be ensure that:

- Every operation falls within the specified time limit
- The periodic (e.g. weekly) maintenance load is as nearly constant as possible

2-2.6: Job Report

This a statement recording the work done and the condition of the equipment. For any planned maintenance scheme to be effective, there must be a continuous flow of information to and from the person(s) doing the work. This feedback is essential for the control and adjustment of the plan. The planner must know exactly what is happening and for that matter, the person who has made the inspection or carried out the maintenance task must report back his findings on the condition of the equipment, the work done and the work still to be done. The report should include information concerning:

- work carried out,

- corrective action taken,
- defects found, corrected and their cause,
- defects observed but not corrected,
- work detailed on the job specification but not carried out or completed,
- major components replaced,
- measurements, of clearances, wear, etc., taken,
- time taken to complete the work, and
- general observations, condition of the equipment.

2-2.7: The history record

The history record is a document on which information about all work done on and / or by a particular equipment is recorded. The success of any system can be evaluated only from the results it achieves. It is upon this evidence that decisions are made for future actions. If the system is to be effective as a controlling medium, it must monitor its progress to check that it is producing the desired result, and assessed, if necessary, the form and extent of corrective measures need. Thus, arrangement for checking re gularly the effect of the maintenance activity must form part of the system.



Self-Assessment 2-2

1. List and explain the subheadings of a planned maintenance scheme that will be used to answer the four broad question in the question 3 of Self-Assessment 2-1.
2. What is a facility register?
3. Maintenance schedule for each equipment consist of an individual sheet, card or a set of sheets/ cards. What should these cards/ sheets indicate?
4. Why is a job report necessary?

Unit 3

TROUBLES AND DEFECTS OF FARM MACHINERY

Introduction

Deterioration of machine parts which results in deviation from their functional capabilities begins immediately after manufacture due to exposure to atmospheric and operation conditions. However, understanding these causes are helpful is formulating preventive measures and solutions to them. Unit 3 introduces the student to machine and machine part failure and its causes.



Learning Objectives

After reading this unit you should be able to:

1. Identify the causes of machine/ machine part failure.
2. Develop a concept of the rate of machine failure over the service life of a machine.
3. Understand what wear is how it occurs.

Unit content

SESSION 1-3: MACHINE FAILURE

- 1-3.1: Major causes of machine failure
- 1-3.2: Basic Definitions
- 1-3.3: The Bathtub Curve

SESSION 2-3: WEAR OF WORKING MEMBERS AND MATING PARTS

- 2-3.1: Types of wear
- 2-3.2: Permissible and Limit Wear (Size)

SESSION 1-3: MACHINE FAILURE

Failure is malfunctioning of the machine. Farm machinery in–service failures can be divided into: (a) destruction – wears, deformations, breakages; (b) maladjustment; (c) clogging with medium handled. Failures can be also classified into abrupt (breakages and the like) and gradual (wear, maladjustments, etc.).

Failure–free performance is the ability of the machine to operate without failures during a certain time period or a certain amount of work performed.

Assets such as engines, tractors and machinery are put into practical use and service to fulfil some specific functions. As a result, failure and deterioration of some of its components is inevitable. Farm machine starts to deteriorate as soon as they are manufactured. Some of the deterioration is due to exposure to air, sunlight, and moisture. Such deterioration could be retarded by storing the machines in a dark shelter. But the most significant deterioration is from the use of the machine.

When the machine is in operation, wear is only one of the processes which cause troubles. While moving along the field a farm machine experiences jolts and jerks and its working members are subject to alternating loads. Dynamic loads applied to machine parts and units loosen attachment and fits of mating parts, displace assemblies on the frame, cause fatigue failure of parts, etc.

When threaded joints get loose, their strength decreases and impact loads rise. This may lead, for instance, to destruction of cutterbars and threshing units, cleaners and the like. Displacement of parts and units because of loosened joints can disturb setting of the working members, axial and parallel alignment of shafts, planeness of chain and belt drive setting. As a result, chains are torn off, shafts are broken, welded joints and frame members are destroyed. Violation of key joint fit stemming from wear of the key or key slot walls is particularly dangerous since it may lead to an accident. If a chain sprocket key is loose on the shaft, the sprocket will move along the shaft and will break down, tear the chain and bend the shaft. Considerable alternating loads deform the frame members, cause cracks and fractures of welded seams.

Deterioration of properties of some materials used in farm machinery is another cause of malfunctions. Parts made from wood, plastics, fabrics, rubber change their physical properties under the action of loads, moisture, sun rays and temperature fluctuations. Conveyor belts loose elasticity, stretch out, weaken, cannot retain setting for a sufficient period of time.

Glands of bearing units, felt collars contacting oil lose resilience and elasticity so that the shielding device becomes leaky admitting abrasives to the bearings of machines.

1-3.1: Major causes of machine failure

Machine failure or breakdown does not happen automatically, just like the way accident happens, they are caused. Machine and equipment failure are largely traceable to the followings:

1. ***Design features:*** There are occasions where aesthetics in design become a trade-off for structural designs, thus such machine fails to meet the required and desired strength to carry out its purpose.
2. ***Age:*** As a machine is put to use, the relative movement of it parts in contact with each other and with the load such as soils in agricultural operations causes wear. This alters the size and shape of machine parts causing them to malfunction and eventually fail. Also, varying load condition induces fatigue which eventually leads to failure. For this reason, older machines are more susceptible to failure than newer ones.
3. ***Environmental factor:*** When machine operates in unpleasant weather and heavy or sticky soil, there is the tendency of machine breakdown or over-bearing of some components, thereby increasing the frequency of component breakdown. Contact with moisture and air from the atmosphere and soil also leads to corrosion which can result in machine component failure.
4. ***Lack of good maintenance practice:*** This factor is interrelated to human factor problems, but differs slightly in that knowledge of good maintenance practice will save the machine from maintenance neglect. Lack of maintenance causes simple solvable problems with a single machine part to affect other parts which leads to machine failure.
5. ***Human factor:*** The single most prevalent factor for machine breakdown and failure is largely traceable to the ***attitude*** of the stakeholders at all levels toward maintenance. For example, negligence of small unpleasant noises and creaking equally leads to major breakdowns.

1-3.2: Basic Definitions

A properly adjusted farm machine operating under specified working conditions should meet a variety of agro–technical and other requirements. The main of these “other” requirements

are reliability, longevity, economy and safety. If a machine meets all the requirements, it is operable and in good repair.

Trouble is a machine condition when even one of the specified requirements is not met. Troubles stem from variations of sizes of working members, mating parts, failure of driving and conveying devices, breakage and deformation of parts and assemblies. There are troubles which do not render the machine inoperable. Deformation of the seed box cover and other minor defects do not disturb operation of the seeder though being in discord with the specifications.

The main causes of farm machine troubles are wear, deformation and breakage of parts and assemblies. To bring such a machine back to operability, it should be subject to repair. Special terms relating both to the machine on the whole and to its individual components are used in operation and repair to define some notions.

Repair is a set of operations for eliminating defects and ensuring safe operation between overhaul periods.

Service life is the machine operation period or the amount of work performed from the beginning of service or after the overhaul till the limit conditions set forth in the specifications. If can also be defined as a duration or scope of work performed by the machine and measured in hectares, tons, hours or other units depending on the machine type.

Longevity is the ability of the machine to preserve its operability up to the limit conditions with the established maintenance and repair system.

Service period is the calendar duration of machine operation since the beginning or after an overhaul and till the limit conditions.

1-3.3: The Bathtub Curve

The bathtub curve, displayed in Figure 3-1, depicts failure rate over the lifespan of all kinds of products be it electrical, electronic, mechanical, etc. which then includes agricultural machinery. Some individual units will fail relatively early (infant mortality failures), others will last until the end of the service life of the machine/ equipment (wear-out), and some will fail during the relatively long period typically called normal life. Failures during infant mortality are highly undesirable and are always caused by defects and blunders: material defects, design blunders, errors in assembly, etc. Normal life failures are normally considered to be random cases of "stress exceeding strength." Wear-out is a fact of life due to fatigue or

depletion of materials (such as lubrication depletion in bearings). A product's useful life is limited by its shortest-lived component. A product manufacturer must assure that all specified materials are adequate to function through the intended product life.

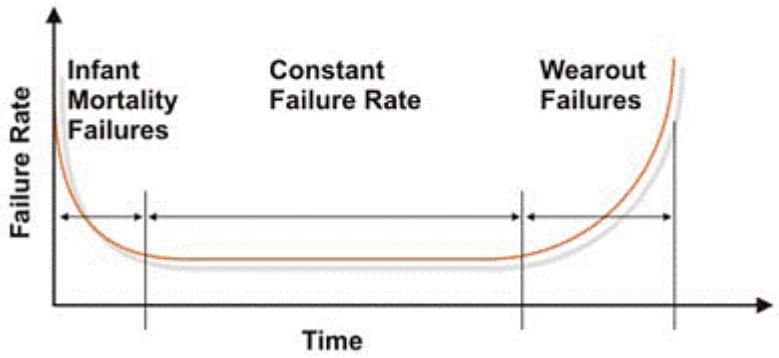


Figure 3-1: The bathtub curve



Self-Assessment 3-1

1. Explain the term failure.
2. What are the major causes of machine failure?
3. Define the following terms as applied to machinery?
    a. Trouble
    b. Repair
    c. Service life
    d. Reliability
    e. Longevity
    f. Service period
4. With the aid of the bathtub curve diagram, explain the failure rate over at various stages over the lifespan of a machine.

SESSION 2-3: WEAR OF WORKING MEMBERS AND MATING PARTS

Wear is a process of destruction and separation of material from the part surface because of friction leading to changes in dimensions, shape and surface condition of the part. It is regarded as the main cause of failures rendering machines inoperable. This process is inherent in all operating mechanisms. It stems from friction forces developing when mating parts are moved relative to one another or when the machine working members interact with the medium handled (soil, vegetation, etc.).

Wear of machine part depends on the material, construction and machine operating conditions. It usually grows gradually without noticeably affecting machine operation for a certain time. Friction of different types is found in mating parts of farm machinery. As regards relative displacement of friction parts, sliding and rolling frictions are distinguished. Friction may be dry, i.e. when no lubricant is provided between friction parts, or lubricated.

The condition of friction surfaces, and lubricant layer between them determine the nature and intensity of friction. The working members of farm machines such as shares, hoes, knives, cylinder bars, blades, etc. operating without lubrication are subject to rapid wear. Wear is a result of wearing measured in established units. The value of part wear is usually determined in linear units (micrometres, millimetres), but sometimes in weight units (milligrams, grams).

Figure 1 represents a graph illustrating relationship between wear and accumulated operating period *T*. The curve consists of three sections: *I* is the run-in period characterized by intensive wear; *II* is the longest period which is characterized by normal operation of mating parts with slow growth of wear and *III* is the rapid wear period when the wear reaches its maximum value and further operation may lead to failure or even to an accident.

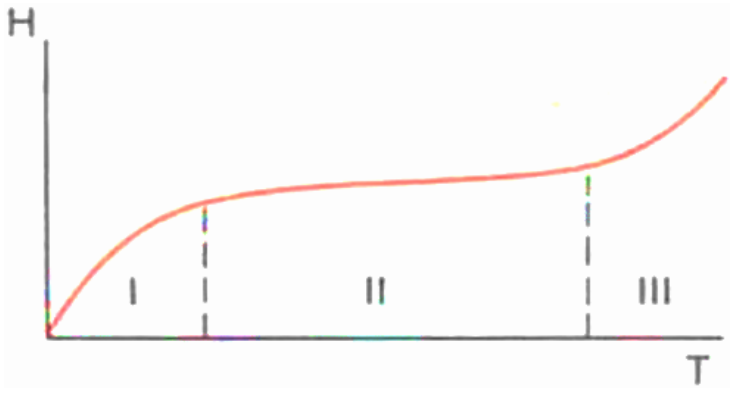


Figure 3-2: Variation of in-service wear of parts

(*I*-run-in wear; *II*-service wear; *III*-accelerated wear)

2-3.1: Types of wear

Mechanical action produces ***mechanical wear***. When rubbing surfaces of parts are moved relative to each other by an external force, a force appears which resists such a movement; it is called friction force. The friction force destroys micro irregularities left on the part surface after machining thus forming abrasion products which are carried away from the friction zone. The wear is maximum at the initial period of operation of a new or overhauled machine. The run-in period is characterized by changes in the part geometry over the friction surface, reduction of friction forces, temperature and intensity of wear.

The condition of the working surface determines the type of wear resistance; therefore the friction surfaces should be finished in compliance with the repair specifications. Solid particles (abrasives) in a free or bonded state cutting or scratching upon the part material produce ***abrasive wear.*** Abrasives are metal particles separated from the base metal as a result of wear, or other particles which get between the rubbing surfaces from some other source, for instance, silicon, quartz, etc. The projections of a solid body or solid particles form recesses (scratches) on the friction surface in the direction of sliding.

The intensity of abrasive wear depends on the quantity, size and hardness of abrasive particles, specific pressure on the friction surface, relative velocities of the rubbing parts and their surface hardness. The harder the metal the better its resistance to abrasive wear. Plough shares, cultivator hoes, tilling and seeding machine disks, knife sections and the like are manufactured from special wear-resistant steels; besides, they undergo thermal treatment, hardening, which considerably increases hardness of their rubbing surfaces.

Abrasive wear is more intensive than wear of other types. It is particularly inherent in parts and bearings which do not have enough protection against ingress of abrasives. To this end various protective devices are used to preclude rapid wear in bearings.

Fatigue wear stems from destruction of micro-volumes of a material surface layer at alternating strain. Fatigue wear is found both with sliding and with rolling friction. Fatigue failure usually begins with cracks which gradually increase in the course of machine operation to terminate in failure. Such troubles occur in reduction gear cases, gearboxes, drum frames, walker bodies, sieve pans, etc. The main features of fatigue failure are as follows: the fracture surface shows no noticeable deformation and the initial crack zone has a smooth

surface and a breakage zone has a rough surface. Tear-off of bolts, flake-like separation of metal and chipping of bearing races and gear teeth leading to pits on the friction surface relate to fatigue failure.

Seizure wear is a result of the seizure of parts, tear-off of base metal, its transfer from one friction surface to another and the action of the irregularities on the mating surface. Friction seizure is local bonding of two solids as a result of high specific loads applied to small contact spots. Separate spots are heated to 1000°C thus becoming fused to each other. The fused spots have no boundaries between the friction surfaces therefore the metal is torn off, and scores in the form of wide and deep grooves appear on the friction surfaces in the direction of sliding. Seizure can even lock the mating parts so that they become stationary.

Oxidation wear is corrosive and mechanical wear in which a chemical reaction between the material and oxygen or oxidizing atmosphere takes the main part. If air humidity is high, minute water droplets settle on metal parts of machines. This moisture reacting with carbon dioxide forms an acid which causes corrosion. The metal surface is gradually destroyed and becomes brown (rust).

Machine assemblies where vegetation residues, dust and other hygroscopic materials usually pile up are particularly liable to corrosion attack. Seeder boxes, fertilizer tanks, bins and facings of combines and other machines made from thin sheet steel are subject to most intensive corrosion. The degree of corrosion failure depends on the air humidity, the duration of machine exposure to rain, thawing snow, etc. To protect metal against corrosion its surface is plated (e.g. with zinc), painted, and in case of a prolonged storage, additionally coated with anti-corrosive compounds.

2-3.2: Permissible and Limit Wear (Size)

The possibility of re-using or the need for repair or replacement of a worn part are determined by measuring its size. The following sizes are distinguished for repair purposes-nominal size, permissible size and limit of size.

Nominal size is the size indicated in the working drawings of a part.

Permissible size is the size (wear) which does not interfere with the normal operation of a machine.

Limit of size is the size (wear) whereat further employment of the part is impossible because of violated agronomic, technical, economic characteristics or hazard of accident. The limit of

wear of the working members of fanning machines is primarily determined by the agro-technical requirements.

The limit of size of a working member is the basic criteria in taking a decision as to discontinuation of work and forwarding the machine for repair.



Self-Assessment 3-2

1. Explain the term wear.
2. Wear of machine parts depend on which factors?
3. Why do farm machine parts such as shares, hoes, knives, cylinder bars and blades experience rapid wear?
4. Explain the 5 types of wear.
5. Define the following terms.
    a. Nominal size.
    b. Permissible size.
    c. Limit of size.

Unit 4

REPAIR OF FARM MACHINERY

Introduction

After a farm machine or its part has failed, it has to be repaired as soon as possible in order to reduce losses incurred due to the machine being out of use. Unit 4 presents to the student what repair means and the organisational setup, processes and materials needed to get a machine or an implement repaired.



Learning Objectives

After reading this unit you should be able to:

1. Define machinery repair and understand the organisational setup and processes for achieving repair.
2. Understand the system of fits, tolerances, accuracy and surface finish associated with machine parts in the repair operation.
3. Identify engineering materials used in farm machinery repair.

Unit content

SESSION 1-4: REPAIR

1-4.1: Repair Organization and Processes

1-4.2: Repair Methods and Organization of Labour

SESSION 2-4: MANUFACTURING AND REPAIR ACCURACY

2-4.1: Part Interchangeability. Fits and Tolerances

2-4.2: Accuracy Classes and Surface Finish

SESSION 3-4: REPAIR MATERIALS

3-4.1: Metals

3-4.2: Woods and Plastics

SESSION 1-4: REPAIR

Repair is a set of operations intended to maintain and restore operability and to renew the overhaul period of a machine. Machines are subject to periodic repairs after certain periods of time (or accumulated amount of work performed). Repair operations include disassembly, washing, troubleshooting, restoration of parts, assembly, adjustment and painting.

Table 4-1: Intervals of maintenance and repair for farm machinery

Description	Intervals, hectares	
	Maintenance	Routine repair
Ploughs	--	300
Disc harrows	--	80
Stubble cleaners	--	220
Cultivators	--	350
Seeders	--	200
Mowers	--	300
Silage combines	60	150
Grain combines	150	400
Corn combines	30	100
Beet combines	20	80
Flax combines	25	50
Potato combines	12	60

In the course of repair, worn-out parts and assemblies are restored or replaced. Depending on the condition of the main units, wear of parts and labour input required to restore the operability and renew the overhaul period, repair is classified into routine and major repairs.

Routine/ Minor repair is intended to ensure or restore operability and consists of replacement or restoration of individual parts. Ploughs, harrows, cultivators, seeders and other relatively simple machines are subject to routine repair. The routine repair of farm machinery except grain cleaners and dryers, animal farm equipment, etc., is usually carried out after the field work season. The repair interval is equal to the annual operation time.

Major repair is intended to restore operability and renew the overhaul period (close to the original one) with the replacement or restoration of any components including the basic ones. The overhaul period close to the original one is established in the relevant technical documents. The basic component is a major part of the machine intended to carry other components. Combines and other complex units are subject to both routine and to major repairs. Major repair intervals for complex farm machines (once in 3 or 4 years) are established according to the operation time and local conditions.

Table 4-1 gives approximate intervals of maintenance and repair for some types of machinery.

1-4.1: Repair Organization and Processes

Production process of a repair unit is interrelated actions of men accomplished in a definite sequence with the use of equipment for restoring operability lost in service and renewing the overhaul period of machines, their units and parts. The production process of a repair shop (enterprise) comprises:

- *Production support*: provision of water, heat, power, compressed air, etc. in the working room, preparation of production means (equipment, fixtures tools, etc.), supply of spare parts and materials;
- *Basic production processes*: fitting, machining, forging, welding and other operations used in repair and assembly of machines;
- *Auxiliary processes*: washing of machines and parts, inspection for defects, running-in of repaired equipment;
- *Production services*: transport, communication, safety measures, fire protection, etc.

Work flow or **process** is a part of the production process intended to qualitatively change the worn machine or its components for restoring the operability and renewing its overhaul period. Mechanical welding, heat treatment and other methods are used to restore the shape, dimensions and other parameters of parts and units in accordance with technical requirements.

When organizing a production process in repair shops and at repair enterprises, the following types of process documents are used:

- instruction cards which include repair operations and specify the machining conditions, equipment used and man-hours required;
- flowsheets which set forth only the most rational sequence of repair operations without indicating machining or process conditions;

- standard practices which contain all necessary operations for eliminating identical defects of constructionally similar parts such as shafts, sprockets, etc.

Operation is a part of a production (repair) process accomplished at one working station. Working station is a part of the production floor area outfitted with all necessary equipment, fixtures and tools.

The stages of the repair process for machines of different types and makes necessarily include external cleaning, washing and disassembly of machines, washing of parts, inspection for defects, restoration of parts and units, assembly, test, adjustment of assemblies, units and machines, run-in, check, painting.

Each of the above-mentioned stages should be based on the most rational practices with the use of modern equipment, tools and fixtures which ensure high quality of repair at lower labour input and cost.

1-4.2: Repair Methods and Organization of Labour

Methods of repair. Farm machines are repaired by individual, flow and unit methods.

The individual repair method is characterized by that the parts and assemblies removed from the machine for repair are not mixed but repaired and mounted back on the same machine. This allows preservation of relative positions of the run-in parts and assemblies. The main disadvantage of the individual repair is a long idle period of the machine since its assembly is possible only after all the parts and units are repaired and checked.

The individual repair method is used in farm shops when the number of machines to be repaired is small.

In *flow* or *indiscriminate repair*, the components go indiscriminate and can be installed after repair onto another machine. This considerably simplifies organization of work and shortens dead periods. The flow repair is used at specialized repair enterprises where a great number of similar machines are repaired.

Unit repair relates to indiscriminate repair at which defective units are replaced with new or pre-repaired ones from the shop repaired stock or from the spare part exchange stations. This method is advantageous as it drastically shortens repair periods.

Organization of labour. Organization of labour in farming equipment repair depends on the volume of repair work performed, variety of machine types, etc.

The simplest type of organizing labour in repair of machines is when the operators themselves repair their machines at the farm shops after completion of the field work. The operators accomplish the bulk of repair work (over 70%) which includes disassembly, washing, fitting, assembly, running-in, etc. without any specialization. Such an organization of labour decreases repair quality, increases consumption of spare parts since the operators try to renew their "own" machines without restoring or reusing old parts. Nevertheless this method is still used in small shops.

Station repair has to do with when the machines and assemblies are repaired at specialized working stations outfitted with necessary equipment, fixtures and tools. One or several repairmen carry out separate repair operations at each station. The specialized stations are organized according to types of machines which are repaired at a given shop.

Apart from repairing large numbers of similar machines, the station organization of labour is used when it is planned to repair machines of different types but in sufficient quantities, for instance ploughs, disc harrows and stubble cleaners, seeders, cultivators, etc. In this case, working stations are organized to perform the following operations:

- disassembly of the machines, repair of frames and running gears, assembly, run-in, adjustments;
- repair of shares, hoes, discs, knives;
- repair of chain and belt drives, conveyor and elevator slats;
- repair of gearboxes, reduction gears, propeller shafts, bearing housing;
- repair of threshing, chopping and cutting drums, beaters, fans;
- repair of shaking pans, sieves.

The station organization decreases time spent for re specialization of workers and employment of special equipment provide for high productivity and good quality of repair. The station organization is used in farm shops and at repair enterprises.

Flow-station organization consists in that units and assemblies are repaired at specialized working stations, whereas the machines (units) are assembled on a transfer line while being moved on a cart along a track or by a special conveyor. The repair and assembly stations are arranged along (or near) the assembly line in compliance with the machine assembly sequence. The flow-station organization is used at specialized repair enterprises. It is the most effective method of repairing great numbers of similar machines.

Depending on the labour input required to perform the entire volume of work one man or a team is assigned to each station. The team leader distributes the work within the team taking into account personal skills. Wages depend on the labour input of each team member. Specialized, integrated and sometimes the so-called "through" teams are organized in repair shops. The specialized teams include workers of one trade, the integrated teams, those of different trades who perform the assigned volume of work (for instance, a fitter, welder and adjuster carry out field repair of machines), while the through teams consist of workers of two or three shifts.

The productive capacity of a shop (enterprise) is the maximum possible number of repaired machines (assemblies, parts) at a complete utilization of the production equipment, floor area, advanced technology and organization of labour. Various measures taken to increase the productive capacity of shops (enterprises) may include increasing the number of shifts, modifying the repair technology, and improving the organization of labour at working places.



Self-Assessment 4-1

1. Explain the term repair.
2. Differentiate between routine and major repairs.
3. What are the stages involved in the repair process for machinery?
4. Define production process.
5. Give the component the following as applied to production process.
    a. Production support.
    b. Basic production processes.
    c. Auxiliary processes.
    d. Production services.

SESSION 2-4: MANUFACTURING AND REPAIR ACCURACY

2-4.1: Part Interchangeability. Fits and Tolerances

Part interchangeability: Manufacturers of farm machinery manufacture machines and spare parts in mass or large-scale quantities. The principal condition of such production is that the machine parts are interchangeable, i.e. any part can be installed without any additional machining or fitting. Interchangeability allows for repairing a machine by replacing old parts with spare parts, often from different manufacturers.

When working out machine drawings, the designer establishes the required manufacturing accuracy for each part. It is hardly possible to make a part strictly to the designed accuracy. Therefore, particular tolerances are indicated in the working drawings for each part. To ensure interchangeability of parts, a system of fits and tolerances is established. These fits and tolerances should be necessarily observed both in manufacture and repair of parts.

System of fits: the following are some commonly used notions and definitions.

Nominal size is a size indicated in the part drawing which is a datum for counting tolerances and determining limits of the size.

Actual size is a size obtained in actual measurement of the part.

Deviation is the difference between the actual and nominal sizes.

Limits of size define the range which should include the actual size. The limit of size may be maximum or minimum and the actual size may be equal to one of them or lie in between.

Upper limit of size is the difference between the maximum limit of size and the nominal size.

Lower limit of size is the difference between the minimum limit of size and the nominal size.

If the limit of size exceeds the nominal size, the deviation is marked with plus (+) in the drawings and technical papers. If the limit of size is smaller than the nominal size, the deviation is marked with minus (-). When one of the limits of size is equal to the nominal size, the deviation is equal to zero and is not marked at all.

Tolerance is the difference between the maximum and minimum limits of size. It is the permissible deviation of a dimension from the specified basic size. It characterizes the established accuracy of dimensioning.

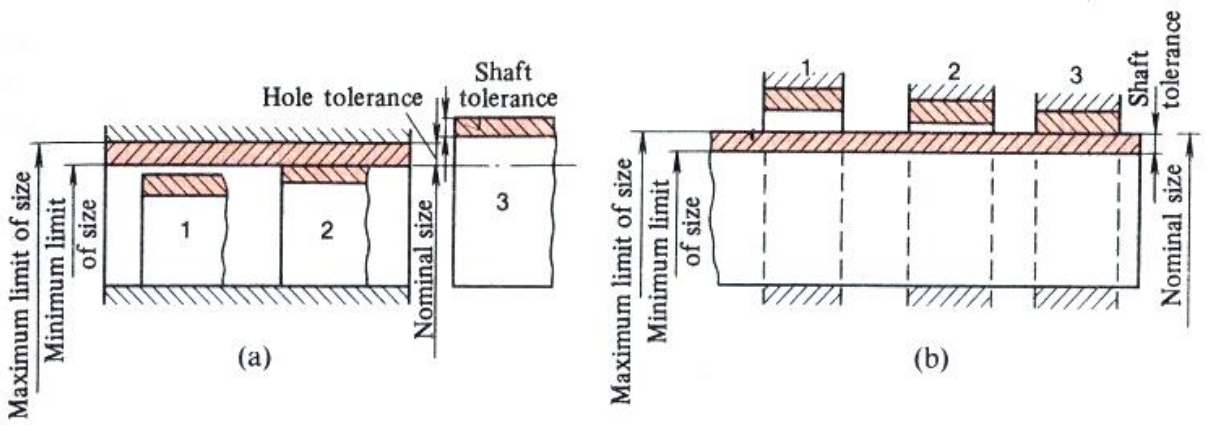


Figure 4-1: Fits and tolerances

(a) Basic hole and (b) Basic shaft systems (1, 2, 3-different limits of size)

Tolerance zone is the range between the maximum and minimum limits of size. If the dimensions are within the tolerance zone, the part is OK, otherwise it should be rejected.

Let us consider the following example. It is necessary to repair a hole whose drawing size is $70^{+0.30}_{+0.10}$ mm. Now the nominal size is 70 mm, the maximum limit of size is 70 + 0.30 = 70.30 mm, the minimum limit of size is 70 + 0.10 = 70.10 mm and tolerance for the size is 70.30 - 70.10 = 0.20 mm.

When two parts are mated with each other, male and female surfaces are distinguished. For cylindrical bodies the female surface is called hole and the male surface shaft. These notions and definitions of cylindrical bodies are also applicable to non-round bodies, such as a key slot and a key, etc.

In assembly of a machine the parts are mated with each other. The joint can be fixed or movable if one part rotates or displaces relative to the other. If a hole diameter is greater than that of a shaft, the difference between them is called *clearance*. If the shaft diameter is greater than the hole diameter, the difference is called *interference*.

The type of the joint of two mated parts, one entering the other, is determined by their fit, i.e. clearance or interference between the shaft and the hole.

Fits and tolerances established for machine building are also used in repair of farm machinery.

The system of fits and tolerances is divided into:

a) basic shaft and basic hole systems according to the basic size;

b) a number of accuracy classes according to tolerance values;

c) a number of fits according to clearance or interference values.

Basic hole system (Figure 4-1) is characterized by the hole having constant size for all the fits of the same accuracy class. Different fits (clearance and interference) are obtained by varying the limits of shaft size.

With all the fits of the basic hole system the lower deviation of the hole is equal to zero, i.e. the minimum limit of size of the hole is equal to the nominal size. Basic hole tolerances are marked with letter A.

Basic shaft system (see Figure 4-1) is characterized by a constant shaft size for all the fits of the same accuracy class and different fits are obtained by varying the limits of hole size. With all the fits of the basic shaft system the upper deviation of the shaft is equal to zero, i.e. the nominal size is the maximum limit of size. Basic shaft tolerances are marked with letter B.

In farm machinery where shafts and axles are often manufactured from drawn steel without machining, the basic shaft system is frequently employed. The basic hole system is also used with many parts and fits of farm machinery. For instance, the antifriction bearing inner races are manufactured in the basic hole system.

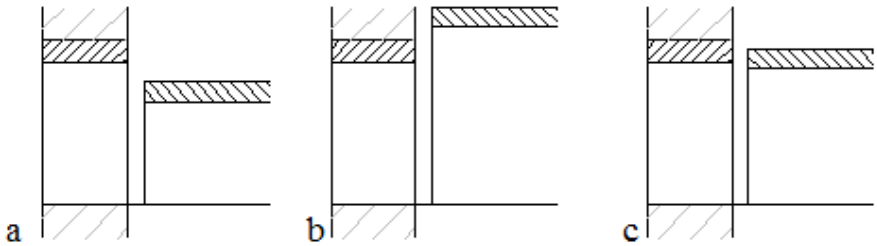


Figure 4-2: a. Clearance, b. Interference and c. Tolerance fits

Fits: this refers to the amount of tightness or looseness between mating parts. There are three types of fits namely clearance, interference and transition fits. *Clearance fit* is one in which there is space remaining after mating parts have been assembled (Figure 4-2a). Clearance fit is established between mating parts of machine that must assemble easily and that must often move relative to one another during normal operation of the machine. Such fits are called running or sliding fits. *Interference fit* is the type of fit that occurs when there is a negative allowance, in other words, when the inside part (e.g. shaft) should be larger than the outside part (e.g. hole) to ensure that mating parts do not move relative to each other (Figure 4-2b). *Transition fit* is the type of fit in which there may be either a clearance or an interference and mating parts have been assembled (Figure 4-2c). This occurs when the smallest shaft size within the allowable tolerance will fit within the largest hole size, resulting in a clearance fit,

and when the largest shaft size within the allowable tolerance will interfere with the smallest hole within an allowable tolerance, creating an interference fit.

2-4.2: Accuracy Classes and Surface Finish

Accuracy classes: Different parts of the same machine can be manufactured or repaired to different accuracies. The technical documents indicate the dimensions and accuracy classes. The accuracy class for the manufacture and repair of parts depends on the requirements to be met by the latter. The sizes ranging from 1 to 500 mm may have one of the following nineteen accuracy classes: 02, 03, 04, 05, 06, 07, 08, 09, 1, 2, 2a, 3, 3a, 4, 5, 7, 8, 9, 10. Classes from 02 to 09 include tolerances for machining high-precision gauge blocks and the like. Farm machinery parts mating with antifriction bearings, gears and some other components are machined to the accuracy class 2, but more often classes 3, 3a and 4. Stamped components and holes for bolts and rivets are to meet the accuracy class 5. When machining non-mating parts or making them by cutting, forging, stamping and rolling, accuracy classes 7, 8 and 9 are used.

Accuracy class 2 is considered to be the primary accuracy class in the tolerance system and therefore the figure 2 is omitted when designating the tolerances of this class. For instance, the designation of the tolerance of the accuracy class 3 basic hole is A_3, while that of the accuracy class 2 is merely A.

When carrying out repair operations, making a new or restoring an old part, always take into account the required accuracy and fits of the assembly to choose the proper method of repair and repair tools.

A designer when designing a machine indicates in a drawing both the part dimensions and the type of fit with mating parts, and the accuracy class. To this end, conventional designation of fit and accuracy class is marked in the drawing. For instance, if size $50A_3$ is indicated for a bearing and $50B_3$ for a shaft, the hole diameter 50 mm should be machined by the basic hole system to the accuracy class 3, while the shaft by the basic shaft system also to the accuracy class 3.

Surface finish: Manual working or machining cannot ensure an ideal smoothness of part surface. The tools leave surface traces in the form of irregularities such as alternating grooves and ridges. Such a condition of the surface is called *roughness*.

In movable fits, excessive roughness of the surface leads to premature wear of parts, since the metal of the ridges (peaks) is subject to abrasion in operation, its particles get into oil thus accelerating wear. In interference fits surface roughness weakens the joint since mutilation of ridges decreases the interference value.

The part surface finish depends on metal properties, tool quality, machining method and conditions. The part service life depends on the surface finish, therefore the latter is indicated in the part drawings alongside with the dimensions.

Surface finish is evaluated in micrometres which are thousandths of a millimetre. Measurements are taken over a surface portion whose length is called sampling length and over a surface whose length is equal to several sampling lengths.

The main parameters used to define surface finish are as follows:

- Average roughness, R_a which is the arithmetical mean deviation of the profile. It could have numerical values ranging from 100 to 0.008 micrometres.
- R_{max} which measures the vertical distance from the highest peak to the lowest valley within five sampling lengths, and selects the highest of the five.
- R_z which averages the five values obtained from the five sampling lengths.

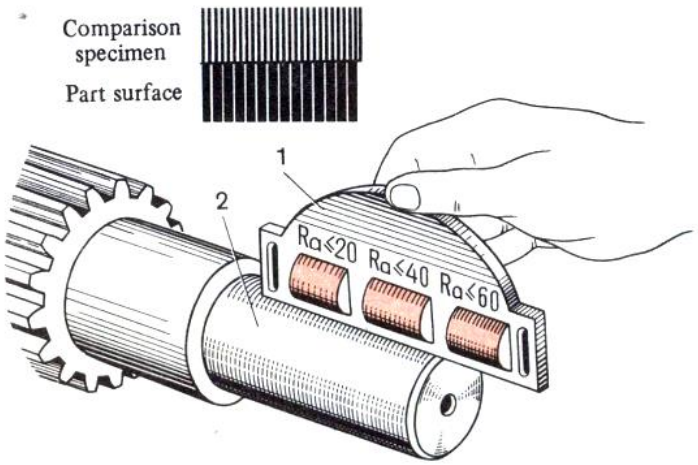


Figure 4-3: Checking shaft surface finish
(1-roughness comparison specimen; 2-shaft journal)

When repairing parts, adhere to their original accuracy and surface finish classes. Surface finish is checked in repair shops with the use of a roughness comparison specimen (Figure 4-3) applied for comparison to the surface to be evaluated.



Self-Assessment 4-2

1. Briefly expound on the idea of part interchangeability.
2. Define the following as applied to system of fits.
    a. Nominal size
    b. Actual size
    c. Deviation
    d. Limits of size
    e. Upper limit of size
    f. Lower limit of size
    g. Tolerance
    h. Tolerance zone
    i. Fit
3. Briefly describe the basic hole system and the basic shaft system and give an example of their application each.
4. Differentiate between clearance, transition and interference fits.
5. Give the relationship between surface roughness and wear and cite 1 example.
6. What are the main parameters used to define surface finish?
7. How should the following parts be machined?
    a. A size of $60A_3$ is indicated for a bearing and 60B for a shaft.

SESSION 3-4: REPAIR MATERIALS

3-4.1: Metals

Repair materials include metals (steel and cast iron), polymers (plastics), wood, rubber, fabric, and the like, i.e. the same materials as used for manufacture of the machines.

Essential metal properties: When using metal for repair, it is necessary to take into account its strength, hardness, ductility, weldability, wear resistance. These properties are determined by tests.

Tensile strength, N/mm^2 is the main characteristic of metal strength. It characterizes the stress corresponding to the maximum load preceding destruction of a specimen relative to its cross-

sectional area before the test. The test is carried out by gradual single-axis stretching of a specimen having definite shape and dimensions.

A tensile strength test also gives the specimen elongation value; the ratio of the elongation to the initial length of the specimen is determined in percent and is called percent elongation (strain). This characteristic is indicative of metal *ductility*. Ductility therefore, is the ability of a metal to deform under tensile stress. It shows the metal's ability to be stretched into a wire.

Malleability refers to a metal's ability to be hammered, pressed or rolled into thin sheets without breaking. In other words, it is the property of the metal to deform under compression. Examples of malleable metals include gold, silver, iron, aluminium, copper and tin.

Metal hardness is a measure of its strength, resistance to abrasion, etc. Metal hardness could be determined through the Brinnel hardness test. This test is carried out by pressing a steel ball of diameter 10 or 5 mm during 10-15 s into a metal specimen at a force P equal to 30000, 10000, 7500, 2500 N. As a result, an indentation is formed on the specimen surface. The softer the metal, the larger the indentation diameter. The indentation diameter is measured with the aid of a magnifying glass and the hardness is found from the relevant table.

In farm workshops, hardness could be checked by filing. Notches left by the file on a metal surface is a measure for approximately determining the metal hardness. When this method is employed, one file should be used for testing and the pressure applied should be similar. To increase the accuracy of the method, select several specimens of metal whose hardness is known beforehand and compare the file notch on them with the notch made on the tested metal.

In repair shops, simplified test procedures such as upsetting, bending and weldability tests are often resort to.

The upsetting test: With this test prepared specimen is upset by a hammer or on a press. If the upset head whose diameter should be twice that of the blank has no cracks or ruptures, the material is suitable for making rivets.

The bending test consists in that a specimen or sheet, strip or bar clamped in a vice and bent through a certain angle, till the specimen ends are parallel or till its halves adjoin each other tightly. The metal is considered to pass the test if bending has caused no exfoliations or cracks.

The weldability test is carried out by welding a specimen. If angle, T-shaped or other rolled section is tested, two specimens are welded along one web. Then the welded joint is broken by striking with a hammer and metal weldability is evaluated by the strength and quality of the weld.

Hot-rolled steel stock: To repair farm machinery, a wide range of hot-rolled bars and sections is used. The rolled stock is made from steels of different grades and comes in different sizes (Figure 4-4).

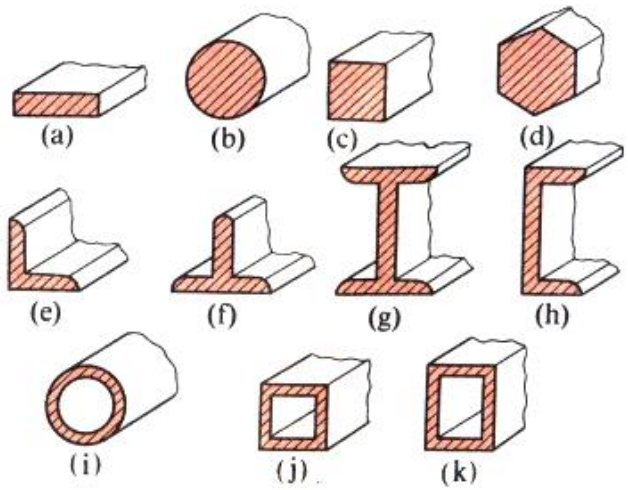


Figure 4-4: Examples of rolled bars and sections
(a-strip; b-round; c-square; d-hexagonal; e-angle; f-T-shaped; g-I-shaped; h-channel; i- pipe; j-square pipe; k - rectangular pipe)

The rolled stock includes strips, round, square and hexagonal bars. The round steel up to 10 mm in diameter is delivered in coils, larger diameter steel comes in bars, 6 and more meters long.

Angle, T-shaped, H-shaped, channel, etc. steel relates to sections. Equal angle bars are manufactured with sizes of 20 × 20 × 3 mm, 30 × 30 × 4 mm, etc. Channel bars have the following section height: 50, 65, 80, 100, 120 mm, etc.

3-4.2: Woods and Plastics

3-4.2.1: Wood

Wood is widely used in farm machinery. Owing to the non-uniform structure of wood, its mechanical properties depend on whether the load is applied along or across the fibres. Wood absorbs moisture from air. Its moisture content is determined by the amount of moisture contained in it related to the weight of wood dried at a temperature of 100-105°C. Moisture

content of fresh wood is 50-90%. Prior to making parts, the wood should be air-dried to moisture content of 12-18%. Therefore, timber arriving at the shop should be subject to drying.

When repairing a machine, broken or worn wooden parts are replaced. The new parts are made from wood of different kinds depending on the purpose, load and operating conditions. Some woods use in farm machinery include pine wood, beech and oak. Oak is superior in strength and longevity to almost all other woods. It is hard, resilient, springy and features low shrinkage. However oak is hard to machine.

Wood bushings arranged at points of friction of farm machinery offer good wear resistance to abrasive media.

3-4.2.2: Plastics

Plastics are produced from resins obtained by processing oil refinery, coke production, etc. waste. Properties of plastics depend on the composition and structure of the initial material, additions (fillers), methods of production; they can be elastic and fragile or hard and soft.

Some plastics offer high strength and low specific gravity (they are 5 to 7 times as light as metals) and exhibit considerable wear resistance. They are readily cast, moulded and machined. Such plastics as fiberglass, capron, etc. are used to make bushings, liners, and other parts. Plastics with wood fillers are employed for making bearings. Electric panels, fittings, insulation are also made from plastics.



Self-Assessment 4-3

1. Define the following metal properties.
    a. Tensile strength
    b. Malleability
    c. Hardness
2. Describe the following simplified test procedures used in the repair shops.
    a. Upsetting test
    b. Bending test
    c. Weldability test

Unit 5

BASIC REPAIR OPERATIONS

Introduction

The repair of farm machinery involves a number of basic operations ranging from the use of hand tools at the bench, to the used of machine tools and welding equipment. Unit 5 identifies some of these operations and the relevant setups, tools and machines used to achieve them.



Learning Objectives

After reading this unit you should be able to:

1. Know what bench work entails.
2. Identify and use basic hand tools used in bench work.
3. Understand the details of operation of a drilling machine.
4. Use the drilling machine for creating holes is workpieces.

Unit content

SESSION 1-5: Bench work

- 1-5.1: Marking
- 1-5.2: Hand chipping (chiselling)
- 1-5.3: Cutting
- 1-5.4: Filing
- 1-5.5: Straightening
- 1-5.6: Abrasion (grinding, sharpening and roughing)

SESSION 2-5: Drilling

- 2-5.1: Twist Drill
- 2-5.2: Drill Features

2-5.3: Drill Operating Parameters

2-5.4: Special Types of Drill

2-5.5: Safety and Care on Drilling

SESSION 1-5: Bench work

Benchwork includes marking, hand chipping, cutting and filing, straightening and bending, drilling and threading, riveting and soldering.

1-5.1: Marking

Marking is the application of lines (scribe marks) and points to the surface of the part under repair to show the limits of metal removal in filing, planing or milling and to lay out future holes. Marking is carried out on a marking-off plate with the use of a scriber, straight-edge, engineer's square, surface gauge, compasses, dividers, trammels, punch and a light hammer (100 g in weight).

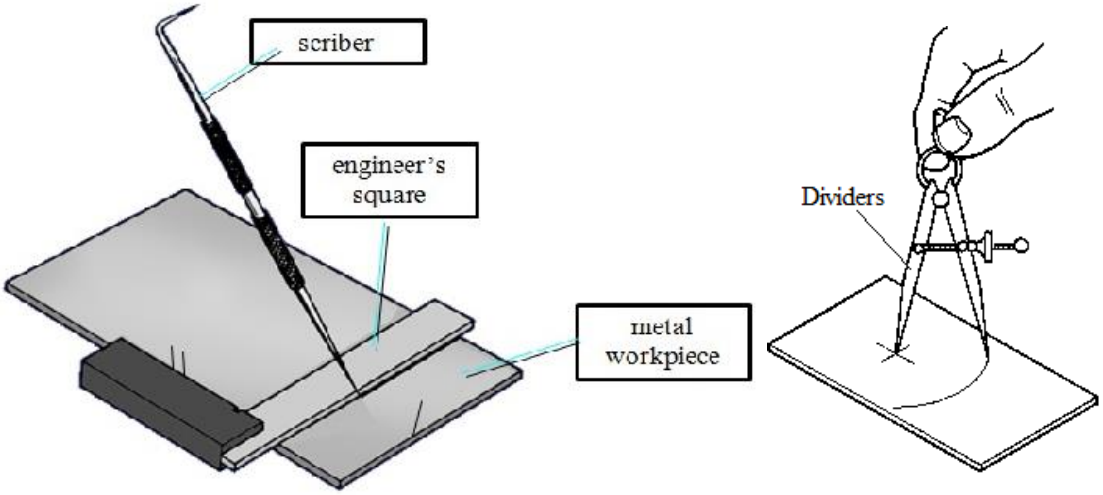


Figure 5-1: Marking out exercises

Correct marking is greatly dependent on properly selected datum (line or surface) from which the sizes are laid off. The machined external edges, axes of symmetry or centre lines applied in the first place are employed as data. Marking is performed against the part drawing or by reference to a new part. Scribe marks are applied to the part surface with a scriber. To preserve the scribe marks to the end of machining and to mark hole centres, small recesses (up to 0.5-1 mm deep) are made with a punch and a hammer. These recesses are called centre pops.

When a great number of similar parts is to be made in repair, a template is prepared and the required dimensions are transferred to the parts with the use of the template.

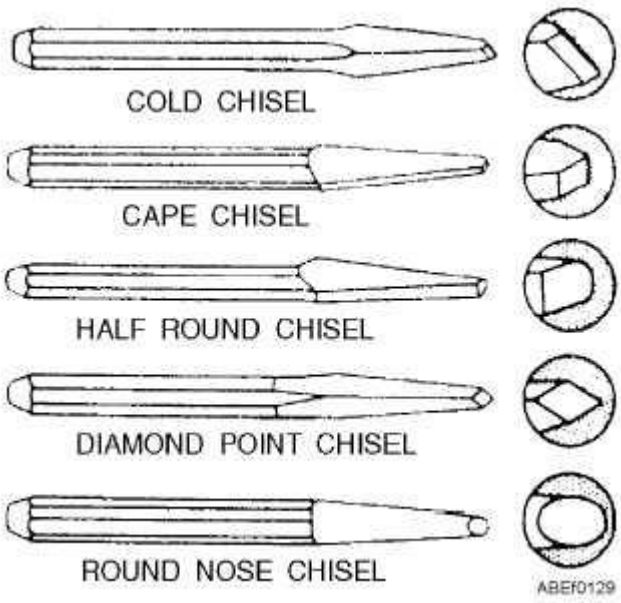


Figure 5-2: Types of chisel

1-5.2: Hand chipping (chiselling)

Hand chipping of sheet, strip, bar metal, etc. is performed in a cold state with the aid of a hammer and flat chisel (or groove chisel). Hand chipping is accomplished by flat chisels having cutting edges from 15 to 25 mm wide. Keyways, oil grooves, etc. are made with a groove chisel. The chisel cutting edge is sharpened on sharpening machines. The cutting edge angle depends on hardness of metal to be worked and is 70° for cast iron, 60° for steel and 45° for aluminium.

To protect workers against chips, the chiselling bench or station should be guarded with a wire gauze or screens and the man who performs chiselling should wear goggles.

1-5.3: Cutting

Cutting is performed in a cold state of metal with the aid of hand shears, hacksaw and crank shears.

Hand shears provide for cutting sheet metal up to 0.5 mm thick. To cut strips, bars and sections, hacksaws are used most frequently. Hacksaw blades come from 0.6 to 1 mm thick and from 320 to 370 mm long. To cut steel, the pitch is 1.6 mm, cast iron 1.4 mm, aluminium

and other soft metals 0.8-1.0 mm. To preclude sticking of the blade, the teeth are set, i.e. adjacent teeth are bent through 0.25-0.60 mm. The blade teeth should be directed forward and the blade should be taut enough.

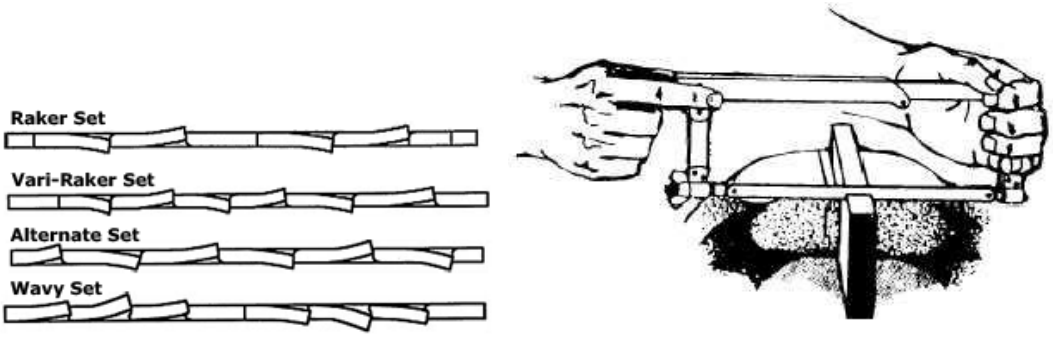


Figure 5-3: Teeth set patterns (left) and hacksaw in use (right)

The hacksaw is moved at a rate of 30-60 double strokes per minute (depending on metal hardness) with such a throw that almost the entire blade length is engaged. When moving the hacksaw forward, it should be loaded, while the reverse stroke should be free.

Power hacksaws are used to cut metal in repair shops.

1-5.4: Filing

Filing is intended to impart the required shape or dimensions to the blank (workpiece) and to fit the surfaces of parts to be mated.

The file should be chosen according to the properties of metal to be worked, shape of the surface (flat, round, oval, etc.) and filing allowance, i.e. metal layer thickness which is to be removed in filing. Bear in mind that the allowance left for subsequent working should be minimum but sufficient to obtain the required shape and dimensions of the part. Files having from 6 to 10 teeth per cm are called coarse files while those having from 12 to 24 teeth per cm are called smooth files. Small files known as needle files are used to work delicate parts. They may also be flat, round, three-square, etc.

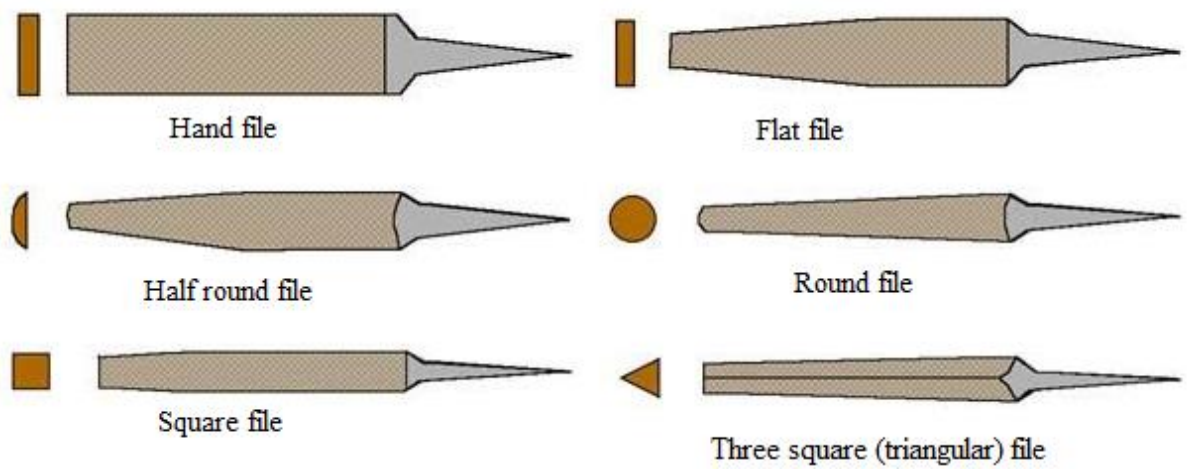


Figure 5-4: Types of files

When filing, the part should be positively secured in a vice and the file handle should be securely fitted. Never use a file with a defective handle or without a handle.

1-5.5: Straightening

Straightening is the elimination of deformation or warpage of a blank (workpiece). Cold straightening is carried out manually on an anvil or cast iron plate with the aid of a hammer. Sometimes, a press is employed for the purpose. Round-head hammers are used for straightening since hammers with square heads indent metal surface. To protect parts with machined surfaces against dents, soft metal or wood cover plates are employed. Sheet metal is straightened by wooden mallets on a plate.

Hot straightening is accomplished after complete or local heating of the part to 850-1100°C.

The deformed members of machine frames are straightened in a cold state by various lever-type fixtures. If the trouble persists, the deformed members are heated with a gas torch and straightening attempt is repeated.

1-5.6: Abrasion (grinding, sharpening and roughing)

Abrasive materials are used to sharpen tools and farm machine parts such as share blades, cultivator hoes, knives, disks, etc. The working member of both sharpening and grinding machines are grinding wheels of different shapes and sizes. The grinding wheels are manufactured from various abrasive materials both natural-such as emery, and corundum- and synthetic materials such as corundum (aluminium oxide-A) and silicon carbide (C). Depending on their size the abrasive material grains are grouped into coarse, medium and

fine, each group having its respective numbers. For example, grain number 40 indicates that the abrasive grain passes through a sieve having approximately 40 meshes to the linear inch.

The abrasive material type and grain size are greatly responsible for surface finish and labour productivity. When manufacturing grinding wheels, abrasive grains are bonded to each other by special binders. The most common bonds used in grinding wheels (with their designations for identification) are vitrified (V), silicate (S), rubber (R), resinoid (B), shellac (E) and oxychloride (O).

The grade of a grinding wheel designates the hardness of the bonded material. A soft wheel is one on which the cutting particles break away rapidly while a hard wheel is one on which the bond successfully opposes this breaking away of the abrasive grain. Most wheels are graded according to hardness by a letter system. Most manufacturers of grinding abrasive wheels use a letter code ranging from A (very soft) to Z (very hard).

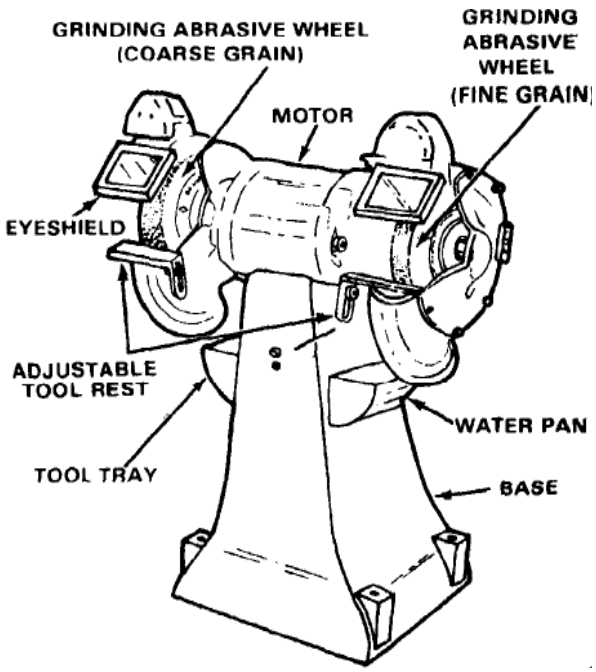


Figure 5-5: Floor-mounted utility grinding machine

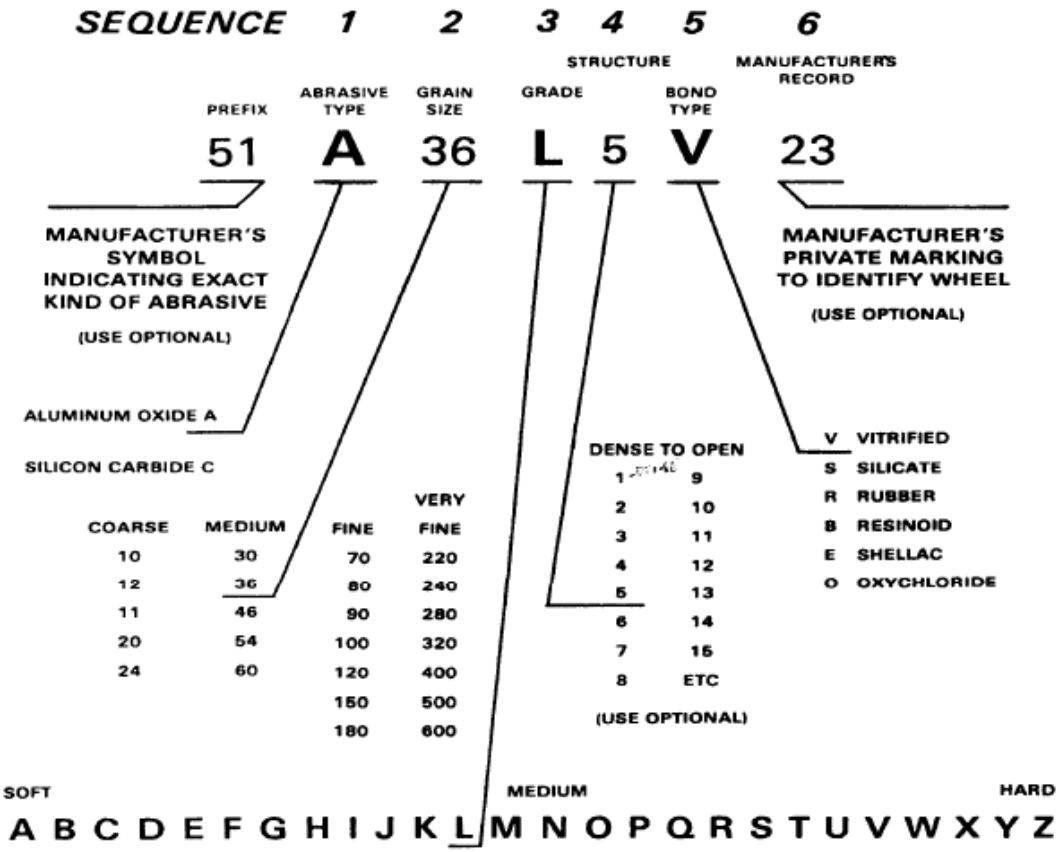


Figure 5-6: Standard system of marking grinding wheels

Bond strength of a grinding wheel is not wholly dependent upon the grade of hardness but depends equally on the structure of the wheel, that is, the spacing of the grain or its density. The structure or spacing is measured in number of grains per cubic inch of wheel volume.

Every grinding wheel is marked by the manufacturer with stencil or a small tag. The manufacturers have worked out a standard system of markings, shown in Figure 5-6. For example use a wheel marked A36-L5-V23. The A refers to the abrasive which is Aluminium oxide. The 36 represents the grain size. The L shows the grade or degree of hardness, which is medium. The 5 refers to the structure of the wheel and the V refers to the bond type.

Prior to mounting on a machine, carefully inspect the grinding wheel and check it for absence of cracks by applying light knocks with a wooden mallet. Do not fail to install washers of a dense cardboard or rubber between the wheel and the mounting flanges.

Abrasive materials are also employed for making abrasive papers and cloths. Depending on the abrasive grain size, the abrasive papers and cloths are marked with appropriate numbers. The larger the number the coarser the grain and hence the rougher the finish.



Self-Assessment 5-1

1. Describe the main benchwork operations used in repair of farm machinery.
2. List the tools used for marking out.
3. List the types of chisels used in the repair shop.
4. List the types of files used in the repair shop.
5. Why are the teeth of a hacksaw blade set?
6. What is straightening as applied in the repair shop?
7. What are abrasive materials used for?
8. What does a grinding wheel with designation A36-L5-V23 stand for?

SESSION 2-5: Drilling

Drilling is the process of cutting holes in metals by using a drilling machine as shown in Figure 5-7. Drills are the tools used to cut away fine shavings of material as the drill advances in a rotational motion through the material.

2-5.1: Twist Drill

The twist drill, as shown in Figure 5-8, is made of high speed steel. It is tempered to give maximum hardness throughout the parallel cutting portion. Flutes are incorporated to carry away the chips of metal and the outside surface is relieved to produce a cutting edge along the leading side of each flute.

2-5.2: Drill Features

The point of the drill is ground to an angle of 59° to the centre line to give two equal cutting edges, and each side is ground back to give "relief" of about 12° to each cutting edge as shown in Figure 5-9.

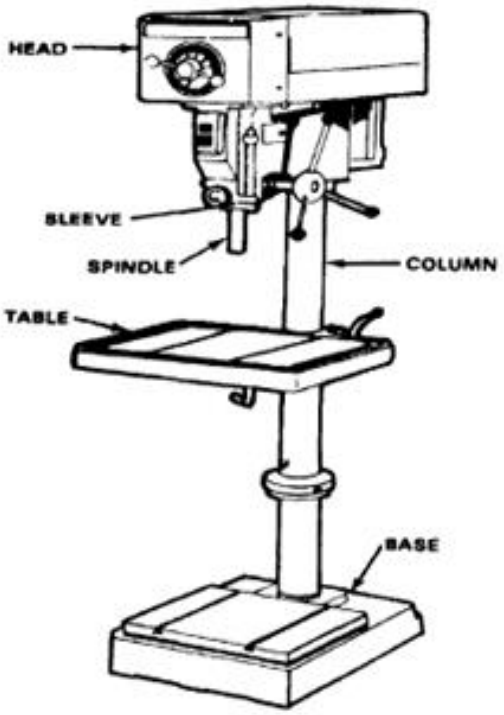


Figure 5-7: Drilling Machine

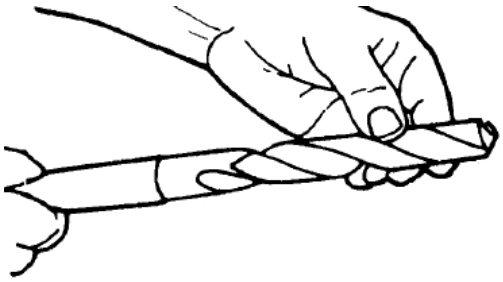


Figure 5-8: Twist Drill

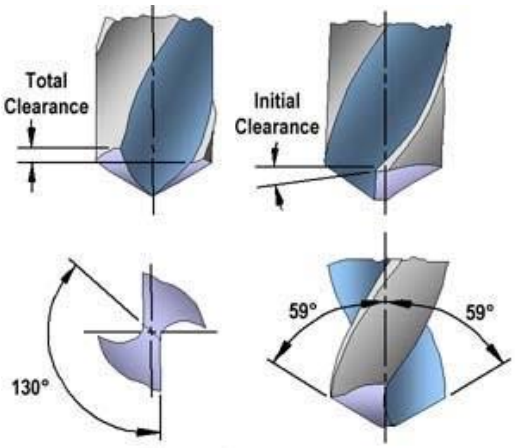


Figure 5-9: Drill Features

2-5.3: Drill Operating Parameters

It is essential to select the correct cutting speed and feed. Tables 5-1 and 5-2 show the most commonly used cutting speeds and feed rates.

Table 5-1: Cutting speed

Material	Cutting Speed
Mild steel	6 - 9 m/min
Stainless Steel	4 - 9 m/min
Aluminium	30 - 36 m/min

Table 5-2: Feed rate

Diameter of twist drill	Feed rate
5.5 mm	0.08 - 0.15 mm/rev
30 mm	0.04 - 0.55 mm/rev

2-5.4: Special Types of Drill

Countersink Drill, as shown in Figure 5-10, is used to form a conical shaped recess to enable a countersunk screw or bolt to fit flush with the surface of the work (Figure 5-11 b and c).

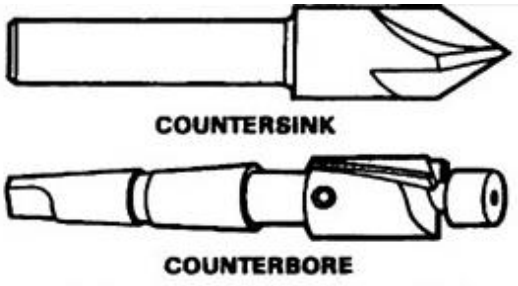


Figure 5-10: Countersink and Counterbore drills

Counterbore Drill, as shown in Figure 5-10, is used to form a flat, or cylindrical recess to accommodate the head of the bolt. It is also used to provide a level base on rough and curved surfaces for nuts and washers as in spotfacing (Figure 5-11a).

2-5.5: Safety and Care on Drilling

i. Twist drill must be clamped in the drill chuck tightly

ii. The workpiece to be drilled must be firmly secured by vice, or clamps.

iii. Drill guard must be closed before switching on the machine.

iv. Use the correct drilling speed and apply suitable drilling force.

v. It is advisable to release the drill occasionally, lift the drill, and clear the hole of cutting.

vi. Apply cutting fluid in the cutting except for drilling cast iron.

vii. Take care, when the drill has nearly penetrated through the workpiece.

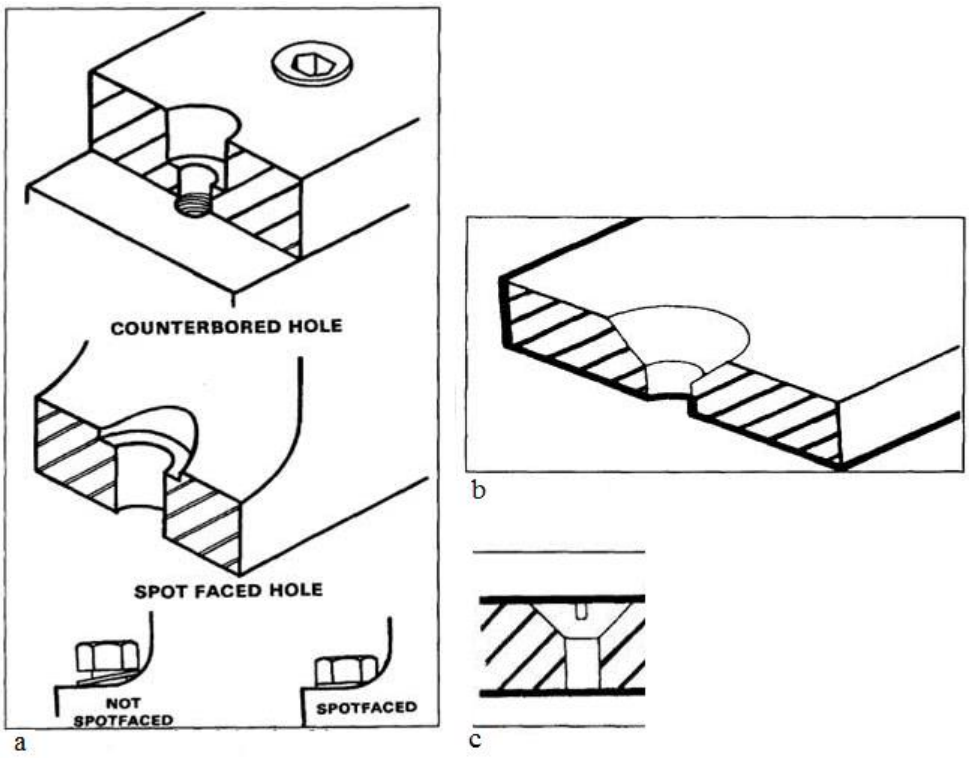


Figure 5-11: a. Counterbored and spotfaced holes, b. countersunk hole and c. countersunk hole with countersink screw

Self-Assessment 5-2

1. Define drilling.
2. Why do twist drills have flutes on them?
3. Give two special types of drill and describe their use with the aid of appropriate diagrams.
4. Which safety measures are to be adhered while drilling?

Unit 6

STRUCTURAL CONNECTIONS

Introduction

Units 6 is a continuation of Unit 5 on some basic workshop operations carried out in the repair of farm machinery. Unit 5 focuses on structural connection operations such as riveting, soldering, brazing, adhesive bonding and welding.



Learning Objectives

After reading this unit you should be able to:

1. Know the various structural connection operations carried out on farm machine parts.
2. Appreciate riveting and design by finding appropriate dimensions, strengths and efficiency of riveted joints.
3. Understand what soldering, brazing and adhesive bonding entail.
4. Know what welding is and identify the types and equipment of welding.
5. Calculate the strength of a welded joint.
6. Carry out welding operation.

Unit content

SESSION 1-6: RIVETING

1-6.1: Hot riveting

1-6.2: Cold riveting

1-6.3: Types of Riveted Joints

1-6.4: Modes of Failure of Riveted joints

1-6.5: Efficiency of Rivet Joint

SESSION 2-6: SOLDERING, BRAZING AND ADHESIVE BONDING

2-6.1: Soldering

2-6.2: Brazing

2-6.3: Adhesive Bonding

SESSION 3-6: WELDING

3-6.1: Plastic welding

3-6.2: Fusion welding

3-6.3: Gas welding

3-6.4: Arc Welding

3-6.5: Types of Welds

3-6.6: Types of Welded Joints

3-6.7: Weld Strength

Maintenance and repair of farm machinery many involve joining or fastening parts of equipment together. Among the common methods of joining and fastening structural components together in the maintenance and repair of farm machinery are:

- Riveting
- Welding
- Brazing
- Soldering
- Bolting

Generally, the method chosen for joining depends on:

1. The service and environment of the component (vibration and corrosion factors).
2. The strength and flexibility requirements
3. Cost
4. The need for disassembly for maintenance
5. Look of finished product

SESSION 1-6: RIVETING

Rivets are used most effectively when rigid fastening is required where a shearing load is expected and when a replacement is needed in future. It may be done hot or cold.

1-6.1: Hot riveting

In this case the steel rivet is heated and while it is red hot it is slotted into the hole and hammered. When the rivet cools it tries to shrink in length, thus holding the joined parts tightly.

1-6.2: Cold riveting

In cold riveting, the rivet is not heated but deformed at both ends until the rivet material fills the hole completely and a new head may be formed at the point end to provide some tensile holding stress. This is more commonly used in the maintenance and repair of farm machinery than the hot riveting.

Damaged rivets are removed by chipping the head and driving out the stem with a punch and a hammer. As depicted in Figure 6-1, the holes in the parts to be jointed are aligned by a punch or a bar after which the new rivet is inserted. The hole prepared to receive the rivet should correspond to the diameter of the hole prepared to receive the rivet and normally be about 0.5 mm bigger than the rivet. The rivet stem length should be sufficient to extend through the parts to be riveted and should protrude for a value sufficient for the head end also called snap head. The rivet stem length l depends on the total thickness t of the parts to be joined and the hole diameter, d:

$$l = t + 1.5d$$

Hand tools, pneumatic riveting hammers and riveting clamps are all used for riveting.

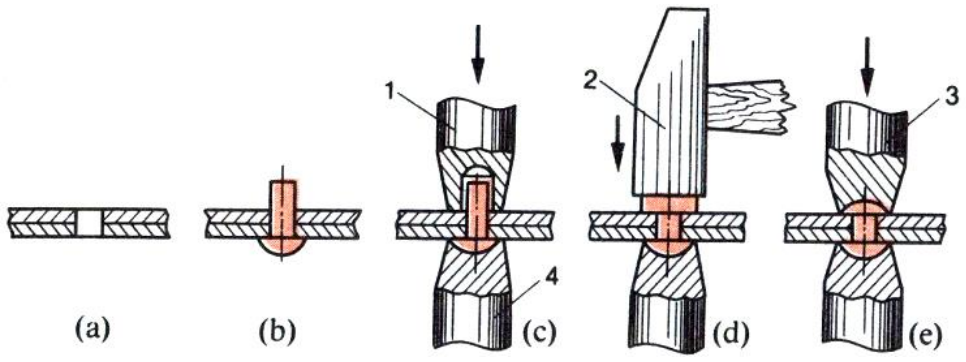


Figure 6-1: Sequence of hand riveting steps

a) aligning holes in sheets; b) inserting rivet; c) upsetting joint by upsetting tool; d) riveting over rivet shank; e) forming snap head; 1-upsetting tool; 2-hammer; 3-snap; 4-dolly

The force needed to separate a riveted joint is the sum of the cross-sectional area of the rivets multiplied by the ultimate share stress of the metal. If there are various possible areas of joint failures, the least value of strength is chosen as the joint strength.

1-6.3: Types of Riveted Joints

Riveted joints are classified as lap joint or butt joint. In a *lap joint*, the edges of the plates are simply laid over each other and riveted as shown in Figure 6-2a. In a *butt joint*, the plates are kept in alignment, butting (i.e. touching) each other and a cover plate (i.e. strap) is placed either on one side or on both sides of the plates. A butt riveted joint with only one cover plate placed on one side is called a *single strap butt joint* (Figure 6-2b), while one in which there are two cover plates placed on both sides of the main plates is called *double strap butt joint* (Figure 6-2c).

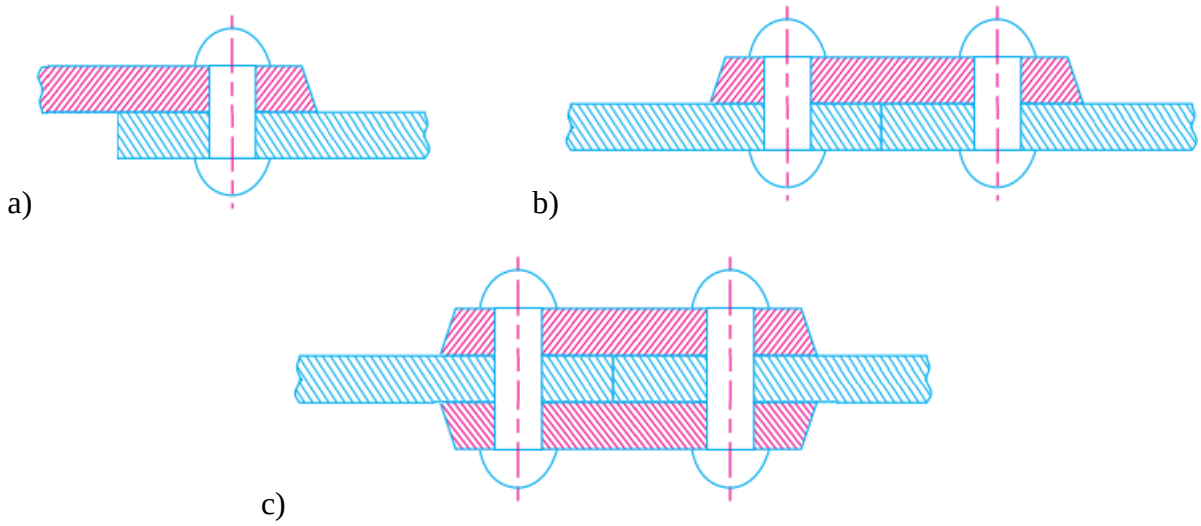


Figure 6-2: Types of Riveted joint.

a) Lap, b) single strap butt joint and c) double strap butt joint

1-6.4: Modes of Failure of Riveted joints

A rivet may fail in the following ways:

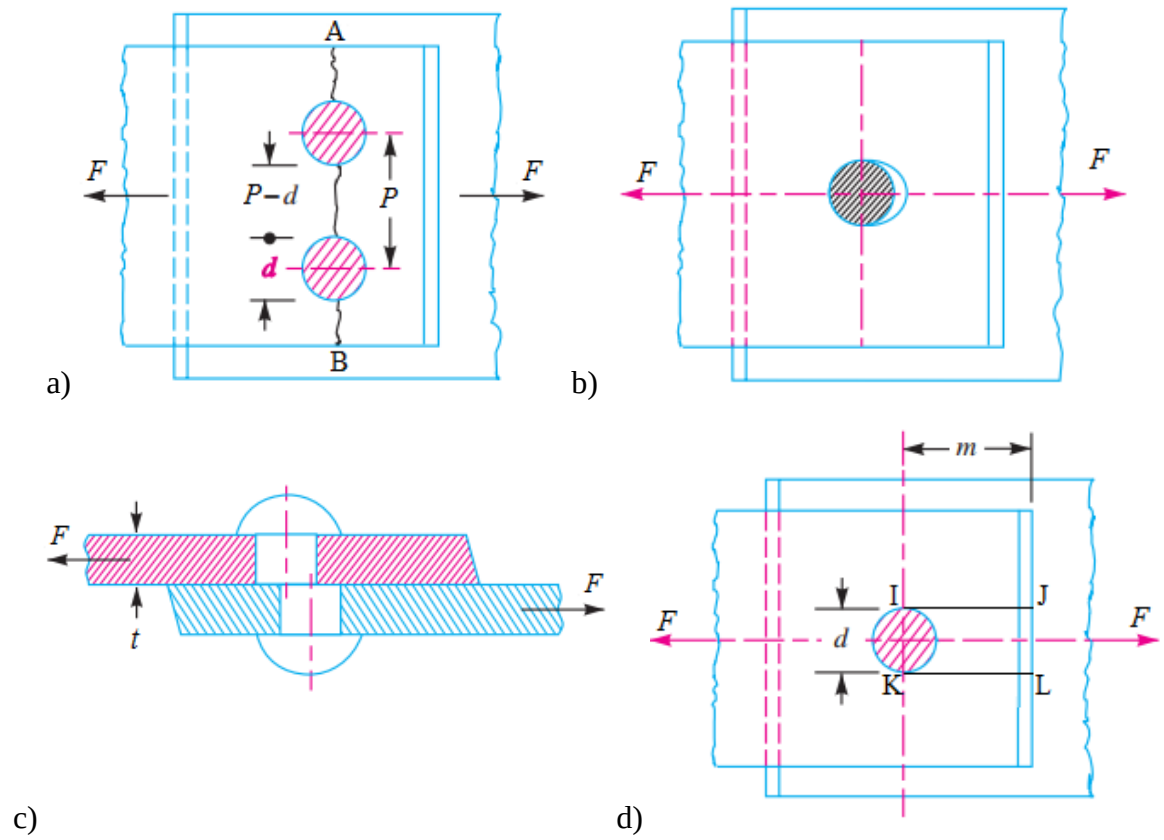


Figure 6-3: Rivet under shear loading

a) Tearing of plate across A-B, b) Crushing of rivet between the rivet and one plate, c) Shearing of rivet and d) Shearing of plate along IJ anf KL.

t = thickness of plates

P = pitch (*Pitch* is the distance from the centre of one rivet to the centre of the next rivet measured parallel to the seam as shown in Figure 6-3a).

m = margin

σ_1 = permissible tensile stress in the plate

σ_2 = permissible crushing/ bearing stress of the rivet or plate material

τ_1 = permissible shear stress in the rivet

τ_2 = permissible shear stress in the plate

There are principal ways in which the joints may fail.

1-6.4.1.1: Tearing of Plate at the Section Weakened by Holes

The plate is obviously stronger at any other section other than section A-B (Figure 6-3a) because section A-B has the least cross-section area. Hence failure by tearing is most likely to

occur at section A-B. If only one pitch length *P* is considered, the joint is weakened by one hole diameter *d*.

$$F_1 = \sigma_1 (P - d)t$$

Where F_1 is the maximum permissible force to tear the plate.

1-6.4.1.2: Crushing of Plate or Rivet

Due to the rivet being compressed against the inner surface of the hole, there is a possibility that either the rivet or the hole surface may be crushed. This causes the rivet hole to become an oval shape and hence the joint becomes loose (Figure 6-3b). The area, which resists this action, is the projected area of hole or rivet on diametral plane which is equal to $d \times t$.

Therefore,

$$F_2 = \sigma_2 (d \times t),$$

Where

F_2 is the maximum permissible force to crush the rivet between the rivet and one of the plates.

1-6.4.1.3: Shearing of Rivet

This failure will occur when all the rivets in a row shear off simultaneously. In a lap joint, failure due to shear may occur only across one section of rivet as shown in Figure 6-3c. However, in case of double strap butt joint, failure may occur across two sections along the stem of the rivet. Shearing area, A_s:

$A_s = \pi \dfrac{d^2}{4}$ for single shear and:

$A_s = 2\pi \dfrac{d^2}{4}$ for double shear.

Therefore, shearing resistance or pull force, F_3 required to shear the rivet per pitch length:

$$F_3 = \tau_1 \left(\frac{\pi d^2}{4} \right)$$

1-6.4.1.4: Shearing of Plates along IJ and KL

As shown in Figure 6-3d, a riveted joint may also fail by shearing in the margin along the planes IJ and KL. This can be prevented by keeping the margin at least $m = 1.5d$.

$$F_4 = \tau_2 (2mt)$$

1-6.5: Efficiency of Rivet Joint

The efficiency of a riveted joint, η is defined as the ratio of the least force/ load needed to cause failure of the riveted joint to the maximum permissible force/ load the parent plates can carry per pitch length. The maximum permissible force/ load the parent plates can carry per pitch length, *F* is given by:

$$F = \sigma_1(Pt)$$

$$\eta = \frac{\text{Least of } F_1, F_2, F_3 \text{ and } F_4}{\sigma_1(P \times t)}$$



Self-Assessment 6-1

1. Which factors are to be considered in choosing a method for joining metals?
2. Whit the aid of a sketch, describe how rivets hold metal pieces together.
3. Find the efficiency of a single riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 50 mm.
   Permissible tensile stress in plate = 120 MP a
   Permissible shearing stress in rivets = 90 MPa
   Permissible crushing stress in rivets = 180 MPa
   (F_1 = 21.6 kN, F_2 = 21.6 kN, F_3 = 28.28 kN and η = 0.6)

SESSION 2-6: SOLDERING, BRAZING AND ADHESIVE BONDING

2-6.1: Soldering

In soldering metal parts are jointed to each other by means of special alloys called solders made of lead-tin alloys. Contrary to welding, the base metal is not heated to plastic state nor melts since the solder melting point is well below that of the base metal. The solder strongly bonds to the base metal and ensures a tight joint.

In repair shops soldering is used for repairing electrical equipment, eliminating leakage of tanks, etc. The area to be soldered is cleaned with a file or abrasive paper, washed and dried. Various etching fluxes are used to degrease the surface and remove oxides and slag from molten solder. Soldering is carried out with an electric soldering iron heated to 300-400°C. Solders have a melting point of 230-270°C.

2-6.2: Brazing

Brazing is a technique of joining metals by fusing a layer of filler rod mostly a brass alloy between the adjoining surfaces. The melting point of the filler (brazing) rod is below that of the base metals being joined but is at a temperature higher than that in soldering. At the correct temperature and with the aid of the flux, the filler rod flows over the heated surfaces bonding to give a joint that in many cases is stronger than the parent metals.

2-6.3: Adhesive Bonding

Adhesive bonding is similar to brazing and soldering but adhesives instead of filler metals are used. There are various adhesives available for tight and strong joining of metals, woods, rubber and other materials in repair. These adhesives include resin-based composition such as epoxy adhesive, etc. These synthetic resin adhesives are used for patching cover plates, mending broken parts and filling in cracks.

Carbinol cement is a thick transparent orange-red liquid. To decrease shrinkage, increase elasticity of joint and accelerate setting, various fillers (iron, cast iron, or aluminium powder, cement, etc.) arc added to the composition.

After applying a layer of carbinol cement the parts are clamped tightly to each other and are kept in such a position indoors at a temperature of 18-20°C for up to 2 days. The exposure time can be decreased to 10-12 hours if using a drying cabinet with a temperature of 50°C.

To ensure strong bonding of parts, the following main requirements should be observed:

- the surfaces should be well fitted to each other;
- the surfaces should be dressed on a grinding wheel, with a file or emery paper;
- the surfaces should be clean, dry and degreased;
- the adhesive layer should be uniformly applied to the surfaces to be joined with well-washed brush in two or three stages depending on the adhesive type;

- the time of drying of the applied adhesive before joining the surfaces and the time of drying of the jointed parts should be strictly observed.



Self-Assessment 6-2

1. Differentiate between soldering, brazing and adhesive bonding.
2. Which requirements should be observed in order to ensure strong bonding of parts?

SESSION 3-6: WELDING

Welding may be defined as a process of joining metal pieces by interpenetration of their atoms. The process is carried out by heating the metal pieces to plastic or fusion state, bringing them together with or without application of pressure and filler material. Apart from joining metals, welding is used in farm machinery repair to stop cracks, build up worn out parts and surface hardening by building up hard alloys on it. Welding processes are classified in two kinds: Plastic welding and Fusion welding. They are also subdivided into different types.

3-6.1: Plastic welding

Forge, resistance welding and thermit welding with pressure. Resistance welding has three types: Butt welding, seam welding and spot welding.

3-6.2: Fusion welding

Gas and Arc welding, and thermit welding without pressure. In order to melt metals to be joined by fusion, a source of high heat energy is required. The two chief methods of obtaining the high temperature required are by a mixture of gases and electric arc which constitute gas welding and arc welding.

3-6.3: Gas welding

The heat for this method of welding is obtained by burning a mixture of combustible gas and oxygen to provide an intense flame. The most widely used combustible gas is acetylene, which when mixed with oxygen gives a flame temperature varying between 3200°C to around

3500°C depending upon the ratio of oxygen and acetylene. Acetylene (C_2H_2) produced by chemical reaction between water and calcium carbide.

The flame is directed on to the metal being joined, resulting in a molten pool of metal. Filler metal is added when required and is usually the same material as that being joined. Fluxes may be required, as in the case of welding aluminium or copper alloys. This method is used to weld a wide variety of metals. It is cheap and portable, making it invaluable for repair work. It can be used on steel up to 25 mm thick, but it is generally employed on sheet material 16 SWG (1.6 mm) and below.

3-6.3.1: Oxy-acetylene flame

The flame produced during gas welding is classified in three groups depending upon the ratio of oxygen and acetylene. These are namely neutral, oxidizing and carburizing flames and are shown in Figure 6-4.

3-6.3.1.1: Neutral flame

It is obtained when equal volumes of oxygen and acetylene are supplied. It has two cones, an inner white cone flame which develops heat for welding and an outer blue flame (enveloping the first) used to pre-heat the metal and protect oxidation. It is used to weld steel, stainless steel, cast iron, copper, aluminium, etc.

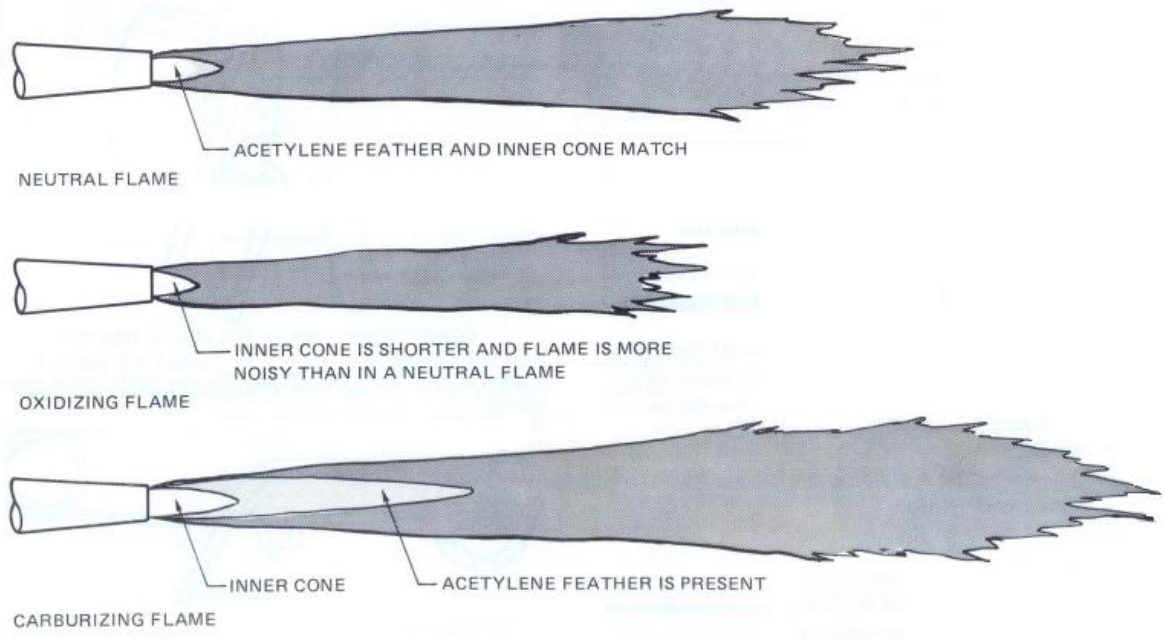


Figure 6-4: Types of oxy-acetylene flames

3-6.3.1.2: Oxidizing flame

It is produced when more oxygen than required for chemical reaction is supplied. In appearance it resembles a neutral flame except that it has a shorter inner white cone flame. It is mostly used to weld brass.

3-6.3.1.3: Carburizing flame

Carburizing flame is produced when more acetylene, than required for chemical reaction is supplied. It has three zones.

- an inner white cone flame
- an intermediate white feather, whitish in colour
- an outer blue flame

It has only a few applications, for example, hard facing, etc.

3-6.3.2: Gas welding technique

The following procedures are adopted to produce a welded joint. The proper size of welding torch and filler material is selected. The welding rod has the same chemical composition as that of the parent metal. The diameter of rod is determined by the empirical formula:

$$d = \frac{t}{2} + 1$$

Where:

d = diameter of the rod (mm)
t = the thickness of the parent metal (mm)

After properly positioning the metal pieces, the acetylene valve is first opened and lighted with a friction spark lighter. The oxygen valve is then opened and the flame adjusted. The joint is the pre-heated. The welding rod is the held in the flame so that its end melts at about the same time as the base metal.

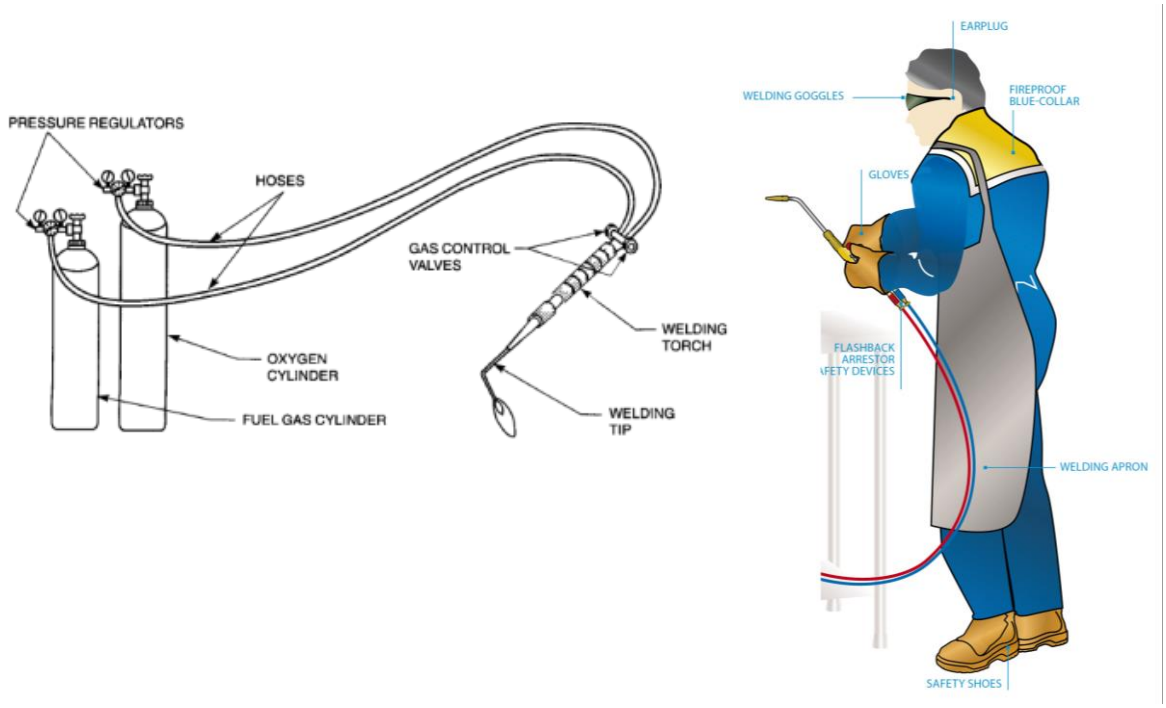


Figure 6-5: Equipment needed for gas welding

3-6.3.3: Equipment needed for Gas welding

Equipment needed to safely carry out gas welding are shown in Figure 6-5 and listed as follows:

- Welding torch
- Pressure regulators
- Hose and fittings
- Gas cylinders
- Goggles
- Gloves
- Spark lighter

3-6.4: Arc Welding

Arc welding is a process utilizing the concentrated heat of an electric arc to join metals by fusion of the parent metals and the addition of metal to the joint usually provided by a consumable electrode. An arc is produced by passing an electric current (either AC or DC) between two electrodes separated by a small gap about 3 to 4 mm. As seen in Figure 6-6, one of the electrodes is the welding rod or wire and the other is the metal plate(s) being welded. The electrodes are connected to an electrical supply to form a circuit. When the two electrodes

touch, current flows and as they are withdrawn, the current continues to flow in the form of spark. The resulting high temperature is sufficient to melt the metal being joined.

3-6.4.1: Welding Terminology

3-6.4.1.1: Filler Material

When welding two pieces of metal together, we often have to leave a space between the joint. The material that is added to fill this space during the welding process is known as the filler material (or filler metal). Two types of filler metals are commonly used in welding are welding rods and welding electrodes.

- **Welding Rod**: The term welding rod refers to a form of filler metal that does not conduct an electric current during the welding process. The only purpose of a welding rod is to supply filler metal to the joint. This type of filler metal is often used for gas welding.

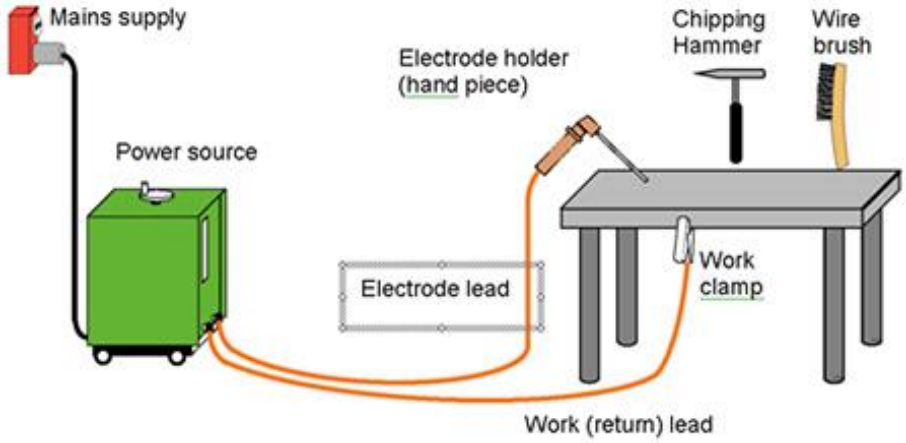


Figure 6-6: A set-up for electric arc welding

- **Welding Electrode**: In electric-arc welding, the term electrode refers to the component that conducts the current from the electrode holder to the metal being welded. Electrodes are classified into two groups: consumable and non-consumable.
    - ***Consumable electrodes*** not only provide a path for the current but also supply filler metal to the joint. An example is the electrode used in shielded metal-arc welding.
    - ***Non-consumable electrodes*** are only used as a conductor for the electrical current, such as in gas tungsten arc welding. The filler metal for gas tungsten arc welding is a hand fed consumable welding rod.

3-6.4.1.2: Flux

Before performing any welding process, the base metal must be cleaned form impurities such as oxides (rust). Unless these oxides are removed by using a proper flux, a faulty weld may result. The term flux refers to a material used to dissolve oxides and release trapped gases and slag (impurities) from the base metal such that the filler metal and the base metal can be fused together. Fluxes come in the form of a paste, powder, or liquid. Different types of fluxes are available and the selection of appropriate flux is usually based on the type of welding and the type of the base metal.

3-6.5: Types of Welds

There are many types of welds. The most common types are the bead, surfacing, plug, slot, fillet, and groove.

3-6.5.1: Weld Bead

This is a weld deposit produced by a single pass with one of the welding processes – Figure 6-7 (a). A weld bead may be either narrow or wide, depending on the amount of transverse oscillation (side-to-side movement) used by the welder. A weld bead made without much weaving motion is often referred to as a stringer bead. On the other hand, a weld bead made with side-to-side oscillation is called a weave bead.



Figure 6-7: Weld Bead and Surfacing Welds

3-6.5.2: Surfacing weld

Several weld beads applied side-by-side are usually used in surfacing which is a welding process used to apply a hard, wear-resistant layer of metal to surfaces or edges of worn-out parts – Figure 6-7 (b).

3-6.5.3: Fillet Weld

A Fillet weld is triangular in shape and this weld is used to join two surfaces that are at approximately right angles to each other in a lap, tee, or comer joint (Figure 6-8).

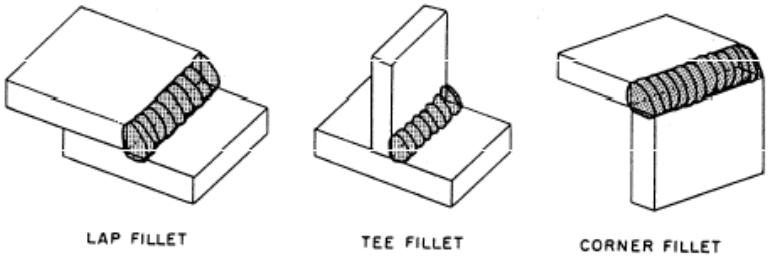


Figure 6-8: Types of Fillet Welds

3-6.5.4: Plug and Slot Welds

Plug and Slot welds are welds made through holes or slots in one member of a lap joint – Figure 6-9. These welds are used to join that member to the surface of another member that has been exposed through the hole.

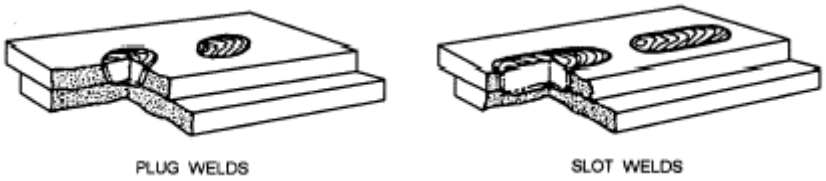


Figure 6-9: Plug and Slot Welds

3-6.5.5: Groove Welds

Groove welds (also may be referred to as Butt welds) are simply welds made in the groove between two members to be joined. The weld is adaptable to a variety of butt joints, as seen in Figure 6-10.

Groove welds may be joined with one or more weld beads, depending on the thickness of the metal. If two or more beads are deposited in the groove, the weld is made with multiple-pass layers. As a rule, a multiple-pass layer is made with stringer beads in manual operations.

The build-up sequence refers to the order in which the beads of a multiple-pass weld are deposited in the joint. Usually, before adding the next pass, the previous pass needs to cool down to a certain temperature which is called the inter-pass temperature. Also, before adding the next pass, the surface of the previous pass needs to be cleaned from slag, especially with shield metal arc welding (SMAW), using a wire brush or other appropriate method.

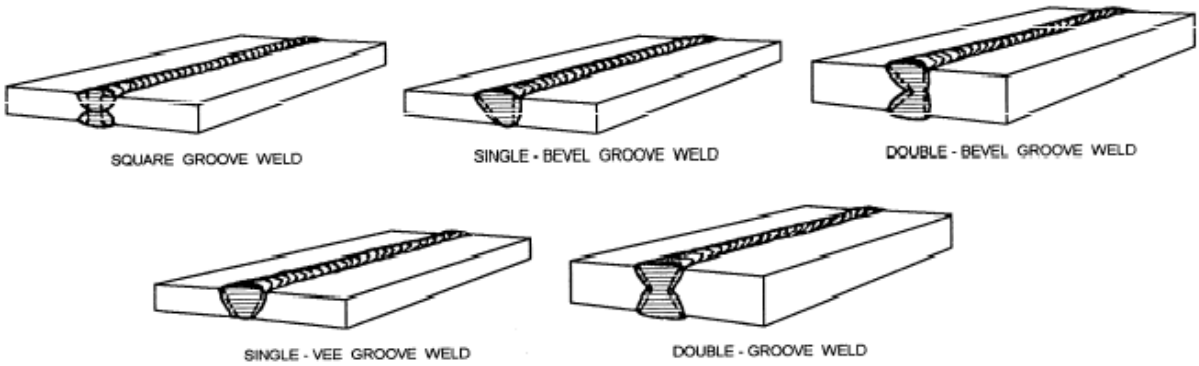


Figure 6-10: Types of Groove Welds

3-6.6: Types of Welded Joints

The weld joint is where two or more metal parts are joined by welding. The five basic types of weld joints are the butt, corner, tee, lap, and edge.

3-6.6.1: Butt Joint

It is used to join two members aligned in the same plane – Figure 6-11 (a). This joint is frequently used in plate, sheet metal, and pipe work.

3-6.6.2: Corner and Tee Joints

These joints are used to join two members located at right angles to each other. In cross section, the corner joint forms an L-shape, and the tee joint has the shape of the letter T - Figure 6-11 (b) and (d).

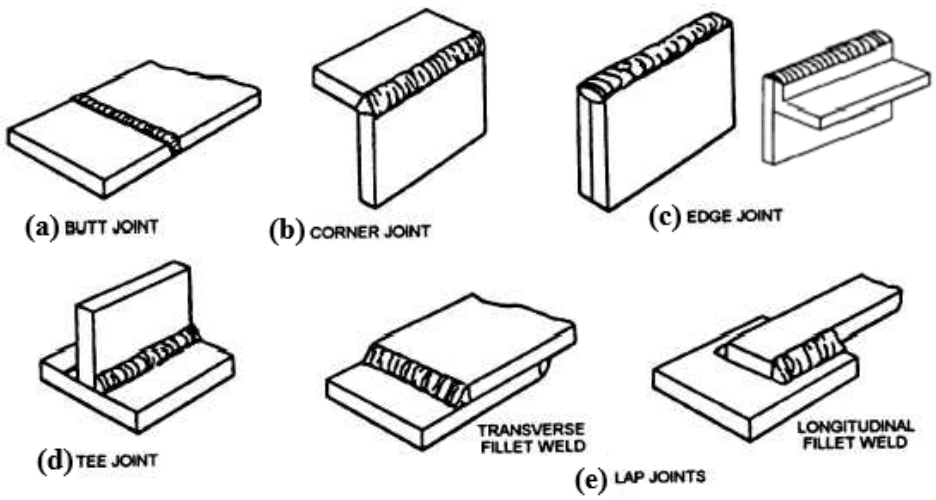


Figure 6-11: Types of Welded Joints

3-6.6.3: Lap Joint

This joint is made by lapping one piece of metal over another - Figure 6-11 (e). This is one of the strongest types of joints available; however, for maximum joint efficiency, the overlap should be at least three times the thickness of the thinnest member of the joint.

3-6.6.4: Edge Joint

It is used to join the edges of two or more members lying in the same plane - Figure 6-11 (c). In most cases, one of the members is flanged, as seen in the figure. This type is frequently used in sheet metal work for joining metals 1/4 inch or less in thickness that are not subjected to heavy loads.

3-6.7: Weld Strength

The strength of a welded connection depends on the base material, the filler material applied, the shape of the weld, the kind of loading and the quality of the workmanship. Generally, the allowable load, *F*, (whether tensile, compressive or shear) is considered to be equal to the area through the weld assumed to be resisting the load (failure plane) multiplied by the allowable working stress, σ.

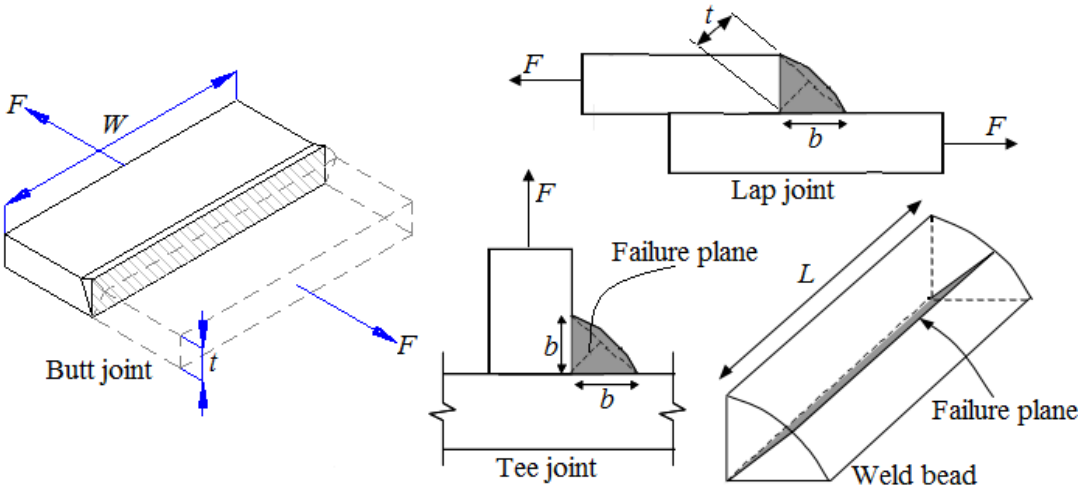


Figure 6-12: Details of butt and fillet welds for strength calculation

For two plates of width, *W*, and thickness, *t*, butt welded, $F = \sigma(t \times L)$

Fillet weld can be performed to create lap or Tee joints. The parameter *b* is called the leg of weld while *L* is the length of weld with *t* (shown on the lap joint) being the throat of weld. Generally, the standard full fillet weld bead has a right angled triangle cross-section such that *b* is made equal to the thickness of the plates. Therefore, the throat $t = b\cos 45° = 0.707b$. The

normal stress is taken to be the shear stress τ and the area assumed to be resisting the weld is always the throat area which is the area of the failure plane. For single weld:

$$F = \tau(L \times t) = \tau(L \times b\cos 45°) = 0.707\tau bL$$

Example 1

Two plates are lapped over each other and joined with two 8 *mm* fillet welds. What length of weld is required for a load of 77.3 *kN* in tension given that the allowable shear stress is 136.7 *MPa*.

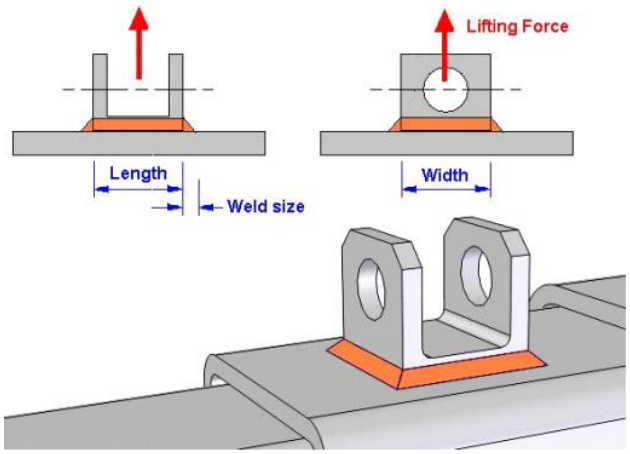
Answer

L = 50 mm



Self-Assessment 6-3

1. Define welding.
2. Describe the set-up and operation of gas welding.
3. Give the characteristics and uses of the three classes of oxy-acetylene flames.
4. Describe the set-up and operation of arc welding.
5. Define the following>
    a. Filler material
    b. Welding rod
    c. Welding electrode
    d. Consumable electrode
    e. Non-consumable electrode
    f. Flux
6. With the aid of sketches, describe the types of welds.
7. With the aid of sketches, describe the types of welded joints.
8. For the figure below, given length = width = 37 *mm*, lifting force is 59 *kN*, and allowable weld stress is 120 *MPa*. Find minimum weld size (same as leg of weld).



Ans: 4.7 *mm*

Unit 7

SPARE PARTS

Introduction

In order to reduce the down time of farm machinery and optimise production, it is necessary to always have a stock of spare parts. However, the ordering of stock should be planned and carried out so as not to overstock (which will lock up some capital making it inaccessible when needed) or understock. The acquisition of this stock is also helped by a series of stakeholders in the supply chain. Unit 7 introduces the student to these concepts and equips them to carry out necessary spare part management decisions.



Learning Objectives

After reading this unit you should be able to:

1. Understand the need to keep stock of spare parts.
2. Know the various stakeholders in the spare part stock and supply chain.
3. Establish an economic order quantity of spare parts.

Unit content

SESSION 1-7: INTRODUCTION

- 1-7.1: Importance of spare parts
- 1-7.2: Spare parts system and supply chain
- 1-7.3: Governments' Role in Assuring Availability of Parts
- 1-7.4: Stock Control
- 1-7.5: Cost of Spare Parts
- 1-7.6: The Economic Order Quantity (EOQ)

SESSION 1-7: INTRODUCTION

A spare part is sometimes called replacement part and it is an additional component part of a machine put aside with the specific objective of using it to replace an existing component when it fails either by breaking, becoming worn out or getting lost or stolen.

1-7.1: Importance of spare parts

The importance of the availability of spare parts for the efficient operation of a workshop should be well recognised.

1. If parts are not readily available, it takes a long time to repair faulty machines.
2. Maintenance operation becomes expensive because of delayed repairs.
3. One of the causes of premature scrapping (write-off) of most agricultural equipment in the country is the non-availability of spare parts.

Careful planning should be exercised continuously, to ensure that needed items can be made available within reasonable time. However, care should be exercised so that the investment in spare parts, stocks and building space, equipment and staff to handle them is kept within reasonable limit.

1-7.2: Spare parts system and supply chain

The objective of spare part system is to ensure that machines suffer minimum delay in being restored to satisfactory operation whenever they break down and that necessary parts are always available to carry out routine maintenance work. The chief factor in minimising delay is local availability of the part and this is best achieved by ensuring the establishment of adequate parts supply system.

The spare part supply system consist of:

1. Network of local stock parts suppliers linked either directly or indirectly to the source of supply. This source may be machinery manufacturers specialising in various types of components, e.g. Mechanical Lloyd for parts of Massey Ferguson Tractor.
2. The local stockist: they stack parts of machinery either on whole sale or at their stores. There are four groups under this:
    i. Franchise group or distributors, e.g. Mechanical Lloyd for Massey Ferguson or the appointed dealers representing the manufacturers.
    ii. Specialist part stockist, e.g. dealers in fan, bearing, etc.

iii. Government Departments
iv. Individual projects

1-7.3: Governments' Role in Assuring Availability of Parts

Governments can play a key role in creating the right environment for the establishment and operation of satisfactory spare parts supply system. This can be done by:

1. According priority in import license allocation (where this is practised).
2. By reducing or waving imports duties for spare parts.
3. By standardisation, e.g. going in for one specific machine and its corresponding spare parts. However, dealers can take advantage and increase prices.

1-7.4: Stock Control

In stock control, the person in charge of store should always monitor inventory figures. The famous 'Saw-curve' stock control is usually applied to manage the stock figures as shown in Figure 7-1. The stock figures depend on the lifespan and stock age of the spare parts, number of equipment, and budget. Planning for procurement is made before the repair parts are reduced to minimum stocks. The planning time depends on the time required to follow through the procurement procedure. The planning is adjusted according to time taken between ordering spare parts and receiving it.

Final stock figures should reduce to zero at the end of equipment lifespan in order to avoid unnecessary expenditure. To manage stock efficiently, an inventory check to confirm correct stock
figures (stocktaking) should be carried out on a regular basis.

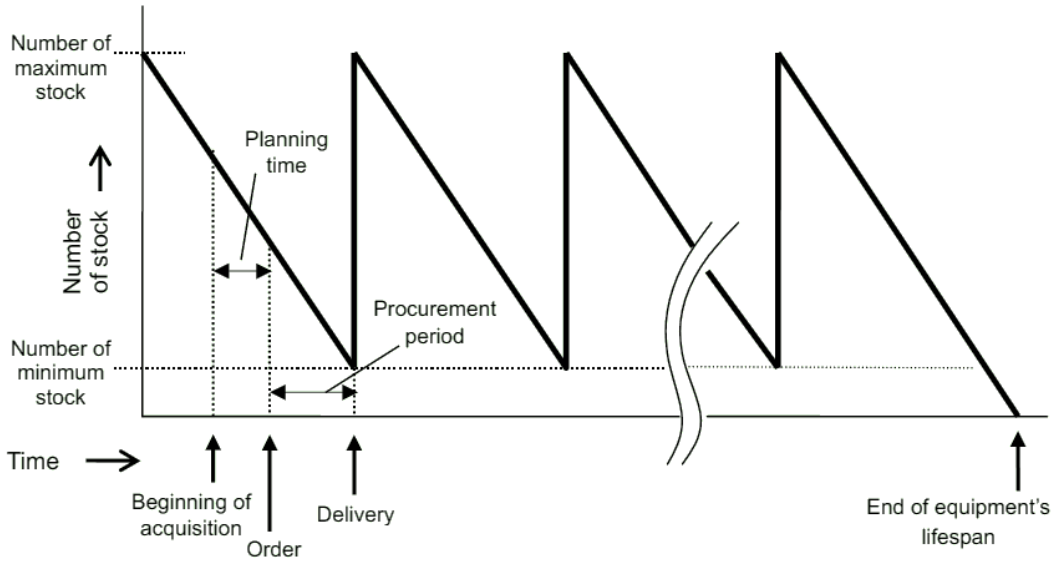


Figure 7-1: The "Saw-curve" spare parts stock control

1-7.4.1: Establishing Stock Levels

Assessment of demand and how to satisfy it economically are the key problems faced in organising a spare part system. Deciding the stock level and the types of parts to be stocked within limits of available investment, storage space and human resources are the major problems. The most important decision in the operation of a spare part supply system is establishing stock levels and maintaining it for individual maintenance at any point in time. The establishment of stock levels is the responsibility of stores and workshop managers. It is their duty to decide on the basis to be followed regarding spare parts procurement and the quantity and variety of stock which should be available. Generally the procurement policy and stock levels to be held will depend mainly on one or more of the following conditions.

1. Proximity to manufacturers or supplier's source and how you can easily communicate with them.
2. The age or state of serviceability of equipment to be serviced.
3. The severity of operators and routine maintenance.
4. The cost of failure to have a part on hand when needed.

1-7.5: Cost of Spare Parts

There are three components:

1. Purchase price (production cost)
2. Holding cost
3. Ordering cost

Holding cost is made up of:

1. capital tied up with the spare parts sometimes called opportunity cost
2. ware housing
3. depreciation of the parts due to obsolesce
4. deterioration
5. pilfering (stealing)

Ordering cost is made up primarily of:

1. Administrative cost and
2. Acquisition cost

1-7.6: The Economic Order Quantity (EOQ)

For large spare parts stores such as those run by dealers, there are various mathematical equations which are used to decide on the optimal or economic order size of spare parts to be ordered occasionally and the level of stock which should be available always. The typical analysis compares the cost of holding and reordering against each other to determine the size of delivery necessary occasionally. When one talks of controlling stocks or inventory, one is concerned with answering questions like:

1. What is the optimum amount of stock to carry
2. What is the optimum economic order quantity for an order from the supplier

There are two methods of deciding the economic order quantity

1. Algebraic method
2. Graphical method

Total cost of stock is the sum of purchase price, ordering cost and holding cost. The most economical point in terms of total stock cost is one where the total cost is the minimum possible.

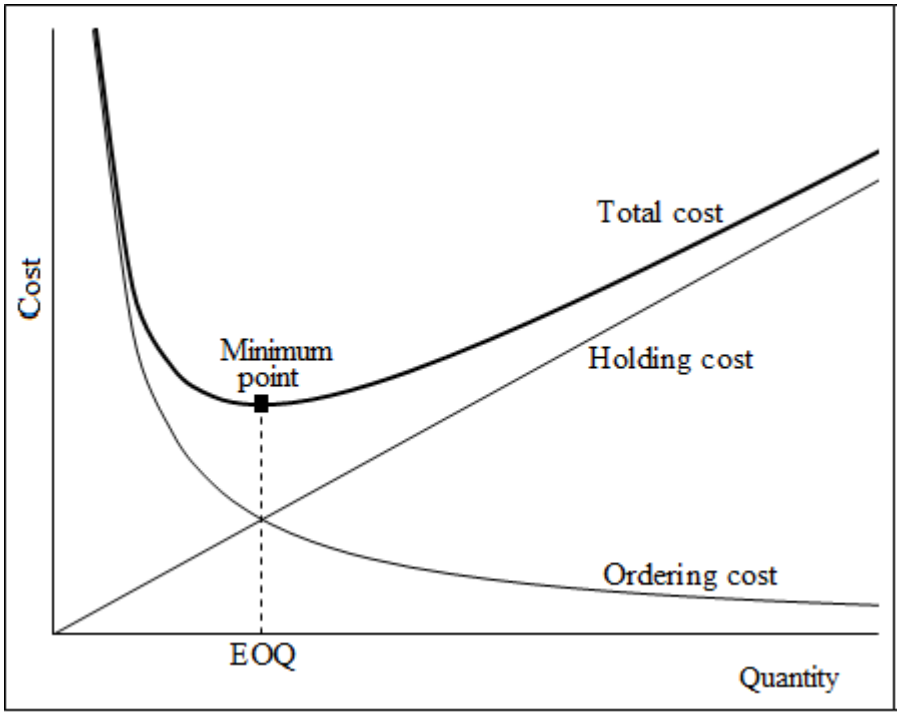


Figure 7-2: Graphical presentation of costs curve

Purchase price = (Part unit cost, D) × (Annual demand quantity, W)

$= DW$

Stock holding cost = (Average quantity in stock) × (Cost of holding one unit of stock)

$$= \left(\frac{1}{2} \times Q\right) \times D \times I$$

Where:

Q = order quantity

D = part unit cost

I = Carrying cost percent $= \dfrac{\text{Annual holding cost per unit}}{\text{Purchase price}}$

Ordering cost = (Number of orders per year) × (Ordering cost per order)

$$= \left(\frac{\text{Total annual demand}}{Q}\right) \times (\text{Ordering cost per order})$$

$$= \frac{W}{Q} \times S$$

Total cost of stock, $TC = DW + \dfrac{QDI}{2} + \dfrac{WS}{Q}$

The minimum possible total cost occurs when:

$$\frac{\partial TC}{\partial Q} = \frac{DI}{2} - \frac{WS}{Q^2} = 0$$

$$Q^2 = \frac{2WS}{DI}$$

$Q = EOQ$ when the total cost is minimum. Therefore:

$$EOQ = \sqrt{\frac{2WS}{DI}}$$

From this model:

a. Minimum total cost $= PD + \frac{EOQ}{2} \times DI + \frac{WS}{EOQ}$

b. Number of orders per annum $= \frac{W}{EOQ}$

c. Length of an order cycle $= \frac{EOQ}{W} \times$ Number of working days in the year

Example

The cost of placing an order for some tractor parts from a supplier in Accra by a part dealer in Ankrakuka is GH¢6. It is estimated that 10000 units of the item is needed in the next 12 months. The price per unit of the items at the suppliers shop is quoted as GH¢24. Given that the holding cost at the dealer's store in Ankrakuka is 2% of the value of the stock. Find the EOQ and the number of orders per annum.

Solution

Ordering cost per order, S = GH¢6

Total annual demand, W = 10000

Cost of a unit item, D = GH¢24

Carrying cost percent I = 0.02

$$EOQ = \sqrt{\frac{2WS}{DI}} = \sqrt{\frac{2 \times 10000 \times 6}{24 \times 0.02}} = 500$$

Number of orders per annum $= \frac{W}{EOQ} = \frac{10000}{500} = 20$

Total cost of stock, TC $= DW + \frac{QDI}{2} + \frac{WS}{Q} = 24 \times 10000 + \frac{500 \times 24 \times 0.02}{2} + \frac{10000 \times 6}{500}$

= 240,000 + 120 + 120

= GH¢240,240



Self-Assessment 7-1

1. What are the effects of the unavailability of machine/ equipment spare parts?
2. What is the objective of establishing a spare parts supply system?
3. What is the role of the government in ensuring the availability of spare parts?
4. What are the components of the cost of spare parts?
5. Derive the formula for determining the economic order quantity of spare parts.

Unit 8

AGRICULTURAL WORKSHOP SAFETY

Introduction

Unit 8 introduces students to safety colours and codes, protection of the body against injury, and safety in the agricultural machinery workshop. This unit also seeks to help student recognize and reduce hazards in the agricultural workshop and react effectively in case of fire or other emergencies.



Learning Objectives

After reading this unit you should be able to:

1. Interpret safety colours and codes.
2. Identify appropriately various protective gear for various parts of the body.
3. Understand the causes, classes and how to extinguish various the various classes of workshop fires.
4. Work safely in an agricultural workshop considering all the above.

Unit content

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1-8.1: Safety Colours

1-8.2: Focal Colours

SESSION 2-8: PROTECTIVE CLOTHING AND DEVICES

2-8.1: Safety Glasses and Face Shields

2-8.2: Protective Clothing

SESSION 3-8: REDUCING FIRE HAZARDS

3-8.1: Causes and Prevention of Fires

3-8.2: Extinguishing Fires

3-8.3: Classes of Fires

3-8.4: Types of Fire Extinguishers

3-8.5: Location of Fire Extinguishers

3-8.6: Using Fire Extinguishers

SESSION 1-8: INTRODUCTION

Work in the farm (repair) workshop involves extensive contact with tools and machinery. As seen in Figure 8-1, this is the area with the highest frequency of accidents in agriculture. For this reason, workers should be especially aware of the hazards that exist and take special precautions as needed. By taking the following precautions a safer work place can be created.

- Install all electrical devices according to the National Electrical Code.
- Install all machinery according to the manufacturer's specifications.
- Keep all tools and equipment adjusted or fitted according to specifications.
- Use tools and equipment skilfully.
- Provide proper storage for tools, materials, fuels, chemicals, and waste materials.
- Keep work areas clean and free of tools, materials, grease, and dirt.
- Keep moving parts properly shielded.
- Manage all situations to avoid the likelihood of falling objects.
- Avoid areas where objects may fall.
- Avoid the flight path of objects that could be thrown by machines.
- Protect eyes, face, feet, and other parts of the body with protective clothing and devices.
- Move slowly enough to avoid creating hazards to self and others.
- Read and heed all precautions.

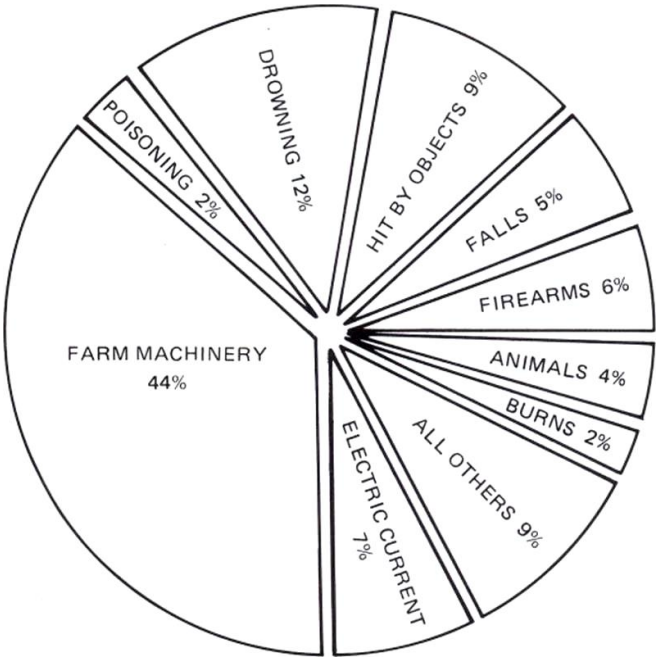


Figure 8-1: Statistics of accidents that occur in agriculture

These precautions help to make the work place safe. However, each person should have insurance to cover any personal injury or property damage that may occur.

1-8.1: Safety Colours

Various organizations have worked together to develop a safety colour coding system for shops. The American Society of Agricultural Engineers and the Safety Committee of the American Vocational Association have published such a code. Colours in the coding system are used to:

- alert people to danger or hazards
- help people to locate certain objects
- make the shop a pleasant place to work
- promote cleanliness and order
- help people to react quickly to emergencies

Each colour or combination of colours conveys a specific message, based upon a standard code. Workshop workers are to familiarise themselves with what each colour stands for in order for the safety message to be understood in the shop. Shops must be properly painted for the colour coding system to make its contribution to shop safety and efficiency.

The following descriptions show how each safety colour is used to convey a safety message and help students work safely in the shop. The colours are used as follows:

Red = Danger. Red is used to identify areas or items of danger or emergency such as s switches and fire equipment.

Orange = Warning. Orange is used to designate machine hazards such as edges and openings. Orange is also used as background for electrical switches, levers, and controls.

Yellow = Caution. Yellow, like the amber traffic light, means be cautious. It is used to identify parts of machines such as wheels, levers, and knobs that control or adjust the machine. Yellow and black stripes are used in combination to mark stairs, protruding objects, and other stationary hazards.

Blue = Information. Blue is used for signs if a warning or caution is intended. Such signs are made of white letters on blue background and carry messages such as "OUT OF ORDER" or "DO NOT OPERATE."

Green = Safety. Safety green is a special shade of green and indicates the presence of safety equipment, safety areas, first aid, and medical practice.

Purple = Radioactivity. Purple is the traditional colour used to designate radiation hazards. However, a black and yellow diagonal striped pattern has been designated as the marking for radiation as of 1986.

White. White is used to mark off traffic areas. White arrows indicate the direction of traffic. White lines also mark work areas around objects in the shop. Yellow may be used in place of white to mark areas and lanes.

White and Black Stripes. White and black in alternate stripes or checkers are traffic markings. An example of such use is to mark barricades which stop traffic.

Gray. Gray is used on floors of work areas in the shop. It is a restful colour and provides good contrast for other safety colours. It is used to paint body areas of machines and may be used on table tops if painting is desired.

1-8.2: Focal Colours

The safety colour coding system includes three focal colours. A focal colour is used to draw attention to large items such as machines, cabinets, and floors. They provide contrast for the

safety colours and create pleasant surroundings for people using the shop. The focal colours are ivory, vista green, and aluminium.

Ivory. Ivory is used to highlight or improve visibility of certain items. These items include tool storage chests, table edges, and freestanding vices and anvils.

Vista Green. Vista green is a special shade of green. It is used to paint bodies of machines, cabinets, and stationary tools such as vices. It is regarded as a pleasing colour and contrasts with the safety colours.

Aluminium. Aluminium is used on waste containers such as those for scrap wood, scrap metal, and rags.

Some of these colours are used in combinations with other colours to mark pipes, hoses, and vents. Such markings identify the material in the lines as oxygen, natural gas, compressed air, air at atmospheric pressure, and water.

A properly colour-coded shop is attractive and pleasant. The colouring system helps guide the user safely through many enjoyable experiences in agricultural mechanics.

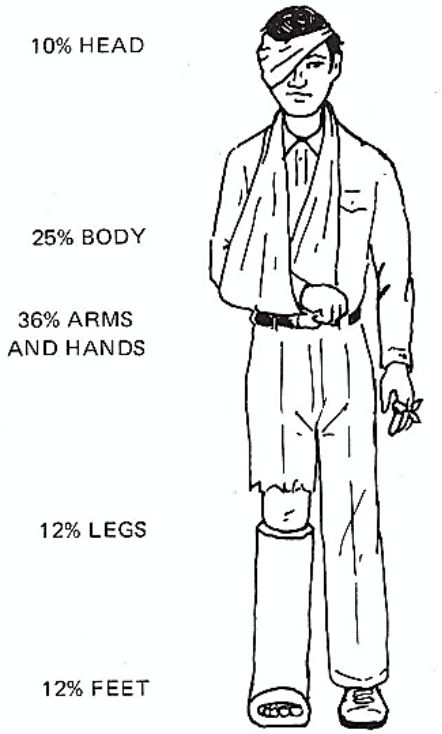


Figure 8-2: Statistics of injuries encountered in the farm workshop



Self-Assessment 8-1

1. What is the area with the highest frequency of accidents in agriculture?
2. Outline the precautions to be followed to ensure a safe work place.
3. What is the purpose of safety colour codes used in the workshop?
4. Match the meaning in column I to the correct colour code in column II.

Column I	Column II
indicates warning	Yellow
indicates danger	Red
indicates information	Green
indicates radioactivity	Blue
indicates caution	Black and yellow stripes
indicates safety	Orange
indicates traffic areas	Black and white stripes
indicates traffic markings	White

SESSION 2-8: PROTECTIVE CLOTHING AND DEVICES

Most work situations in agricultural machine shops require some type of body protection. The eyes, ears, hands, arms, feet, and legs are easily injured, Figure 8-2. The best protection against injury is to prevent the accident from happening. The next best approach is to protect the body where injury may occur. Both lines of defence are needed to minimize injuries.

2-8.1: Safety Glasses and Face Shields

The face and eyes are regarded as the most critical parts of the body to be protected. This is because the eyes are so easily damaged and the face is easily disfigured. Flying objects striking the head can easily cause blindness or result in death. Acids, caustic chemicals, fertilizers, pesticides, solvents, molten metal, and hot water are all dangerous materials. Those who work around these materials should shield themselves against the possibility of accidental contact with them.

Safety Glasses and Goggles: Safety glasses and goggles offer minimum eye protection and are the first line of defence for the eyes, Figure 8-3. Glasses and goggles should be the approved type with special impact-resistant lenses and side shields. They should fit the face and be kept clean for proper visibility. Single-piece goggles and cup goggles are used when special eye protection is needed against chemicals, flying objects, or damaging light. Special shaded lenses are needed when welding.



Figure 8-3: Safety glasses, goggles and face shield

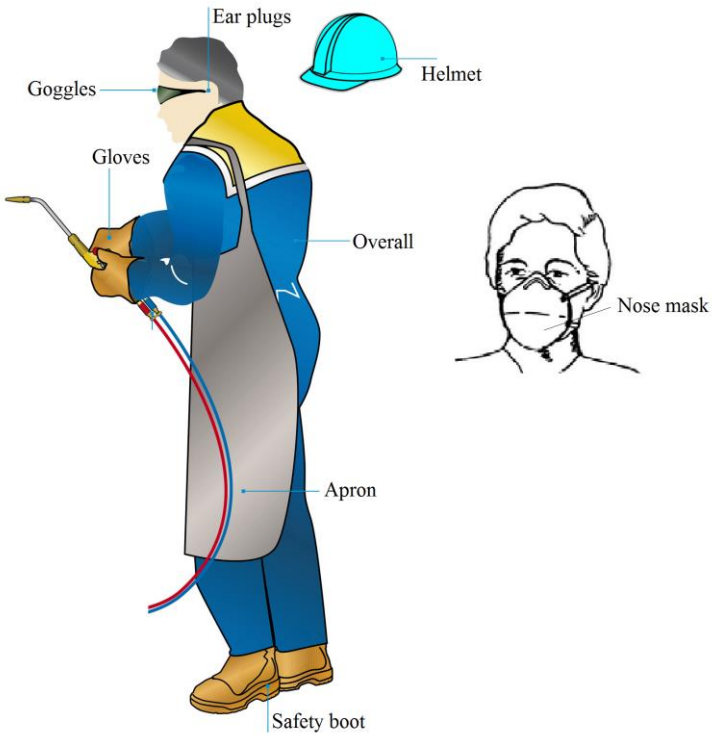


Figure 8-4: Protective clothing and devices

2-8.2: Protective Clothing

Suitable protective clothing that fits properly helps to prevent or reduce injuries. There should be no cuffs, strings, or ties that could be caught by turning machinery. Clothing should be fire resistant and provide protection from scrapes and abrasions. Protective clothing must be easily cleaned. Keeping clothing clean keeps clothing more fire resistant.

Overalls: Overalls are the most versatile and all-around item of clothing for agricultural machine shop users. They protect the arms, body, and legs. They can be buttoned or zipped to the neck for maximum protection. Overalls should fit well and be easily removable. They have the advantage of many pockets for pencil, pad, small tools, and objects which the worker uses.

Aprons: Heavy cloth or leather aprons provide good protection for the front of the body and upper legs. Aprons are recommended only for limited shop work at benches or as additional covering over overalls.

Safety boot: Leather shoes with steel toes are recommended when working in the shop and when using machinery, Figure 8-4.

Gloves: These are used to protect the hands from excessive abrasion, heat, liquids or chemicals. Leather gloves are needed where protection from heat, abrasion, or impact is needed. Only rubber or vinyl gloves are suitable where liquids or chemicals are involved.

Helmets (Hardhats): Helmets are needed when working where objects are above head level or flying objects could be encountered. They are made from special lightweight and impact-resistant materials.

Nose Masks: Masks that cover the nose and mouth are needed to filter out particles of dust or spray paint. Such materials irritate the nostrils and sinuses. Continuous inhaling of dust leads to lung diseases such as black lung and cancer. Effective dust masks are not expensive and should be worn when sanding, painting, welding, mixing soil, shovelling grain, or whenever dust is encountered.

Table 8-1: Duration of time permitted at various sound levels

Duration per day in hours	Sound level (dB)
8	90
6	95
3	97
2	100
1½	102
1	105
½	110

¼ or less	115

Ear Muffs and Ear Plugs: Wearing of ear muffs and plugs for protection is recommended when working in certain types, levels, intensity, frequency and duration of noise. Equipment such as the radial arm saw, planer, chain saw, tractor, and lawnmower can produce noise that may damage hearing. *Noise intensity* refers to the energy in the sound waves. *Noise duration* refers to the length of time a person is exposed to a sound. Duration of exposure to various sound levels recommended by the Occupational Safety and Health Act (OSHA) of the US are shown in Table 8-1. Decibel (dB) is the standard unit of sound.



Self-Assessment 8-2

1. Which parts of the body are easily injured?
2. List all the safety devices worn in the workshop and indicated their function.
3. What are the permissible duration of exposure to various sound levels established by the Occupational Safety and Health Act (OSHA)?

SESSION 3-8: REDUCING FIRE HAZARDS

3-8.1: Causes and Prevention of Fires

To produce fire, three components must be present at the same time. These three components are fuel, heat, and oxygen; they are known as the fire triangle. If any one of these three components is eliminated, fire will be prevented from starting, or it will be stopped if it has started. Therefore, in order to prevent, control, or stop fires:

- store fuels in approved containers
- store fuels away from other materials that burn easily
- store materials in areas that are cooler than their combustion temperature
- use fire only in safe surroundings
- put out fires by removing one or more elements in the fire triangle

The prevention of fire goes hand-in-hand with safe use of equipment and efficient management of work areas. For instance, the proper use of a gas cutting torch decreases the likelihood of fire resulting from its use. Proper storage of materials decreases the chance of fire and keeps materials readily available when needed. A clean work area also decreases the chance of a fire. Special paint booths provide a clean area for paint jobs and also decrease the likelihood of fire.

3-8.2: Extinguishing Fires

Fires are extinguished or put out by adding water to cool them, covering them to cut off the oxygen, or by removing the fuel. This could mean wrapping a person whose clothing is on fire with a blanket. It could mean stopping fire in a field by raking grass and leaves out of the path of the fire or by throwing soil on the fire to smother it. Fire at a gas torch or hoses may be stopped by shutting off the gas at the cylinder. A burning container of paper may be extinguished by cooling it with water from a hose or bucket.

3-8.3: Classes of Fires

To effectively and safely put out a fire with a fire extinguisher, the class of fire must be known. This is determined by the material and the surroundings as follows:

- *Class A* - Ordinary Combustibles. Ordinary combustibles include wood, paper and trash. Class A combustibles do not include any item in the presence of electricity or any type of liquid.
- *Class B* - Flammable Liquids. Flammable liquids include fuels, greases, paints, and other liquids, as long as they are not in the presence of electricity.
- *Class C* - Electrical Equipment. Class C fires involve the presence of electricity.
- *Class D* - Combustible Metals. Combustible metals are metals that burn. These metals include magnesium, potassium, titanium, and zirconium. Burning metals are very difficult to extinguish. Only Class D extinguishers will work on burning metals.

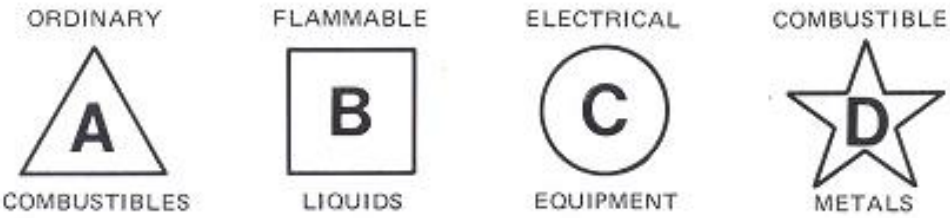


Figure 8-5: Standardised symbols on fire extinguishers indicating the class of fire they are used for

Fire classification is based on how to safely and cheaply extinguish each type of material. Water is generally the cheapest material to use in fire control, but it may not be safe or effective. A firefighter can be electrocuted if the stream of water hits exposed electrical wires, plugs, appliances, or controls. Water is not suitable on fires involving petroleum products since the fuel floats to the top of the water and continues to burn.

3-8.4: Types of Fire Extinguishers

The proper fire extinguisher can put out a fire within seconds. However, such results occur only if the fire is extinguished when it first bursts into flames. The key is the proper extinguisher, used immediately, and in the proper way. This combination may make the difference between a mere frightful moment or a multimillion dollar fire loss with serious injuries and death.

Common types of extinguishers include:

- water with pump or gas pressure (used for Class A fires)
- carbon dioxide gas (CO_2) (used for Class B and C fires)
- dry chemical powders such as purple-K (used for Class A, B and C fires)
- Dry powder agents for Class D fires work by smothering and heat absorption. The most common of these agents are sodium chloride granules and graphite powder. In recent years powdered copper has also come into use.
- foam (used for Class A and B fires)
- blanket (used for smothering fires on humans or animals)

Extinguishers are marked according to the class or classes of fires on which they will safely work. Fire extinguisher labels contain standardized symbols to help the reader act quickly in an emergency. The symbols are as follows:

- *Green triangle*—for Class A, ordinary combustibles

- *Red square*—for Class B, flammable liquids
- *Blue circle*—for Class C, electrical equipment
- *Yellow star*—for Class D, combustible metals

3-8.5: Location of Fire Extinguishers

The location of fire extinguishers is very important. A few seconds lost in looking for the right type of extinguisher could allow a fire to rage out of control. Extinguishers for each class of fire should be placed in areas where their respective classes of fires are likely to occur.

Extinguishers should be placed in clean, dry locations near exits within easy reach. The extinguisher should be hung on the wall so that the top of the extinguisher is not more than 1.5 *m* above the floor. The bottom of the extinguisher should be at least 10 *cm* above the floor.

3-8.6: Using Fire Extinguishers

Generally, fire extinguishers are held upright and operated by a lever. However, some types are activated by inverting the tank, causing chemicals to mix inside the container. Sometimes the lever is blocked by a pin to prevent accidental discharge. Before operating an extinguisher, the instructions on the container should have been read. For most extinguishers, a pin is pulled and a lever pressed.



Figure 8-6: Instruction for operating fire extinguisher

Before discharging an extinguisher, the firefighter should move to within 2 to 3 *m* of the fire and direct the extinguisher nozzle toward the base of the fire, Figure 8-6. The extinguisher will be empty in a matter of seconds, so any material that misses the fuel at the base of the fire will be wasted. A monthly inspection of all fire extinguishers should be made to assure that the extinguishers are useable in case of an emergency.

Always think before acting. Call for help immediately. Be sure that any fire is completely out before leaving the area. The local fire department is the best source of help on fire safety and prevention.



Self-Assessment 8-3

1. What are the components of the fire triangle?
2. Which measure are to be put in place in order to prevent, control, or stop fires?
3. Describe the following classes of fire and give the types of fire extinguishers needed to quench each of them.
    a. Class A
    b. Class B
    c. Class C
    d. Class D
4. March the symbols used on fire extinguishers to the classes of fire they are used against.
5. Outline the procedure for using a fire extinguisher.